INDEX

NAME: Vijay Antony C STD: _____ SEC.: _____ ROLL NO.: 24180131

S.No.	Date	Title	Page No.	Teacher's Sign / Remarks
1	25/07/26	Study of various Network commands used in Linux & Windows		
2	1/08/26	Experiment on CISCO packet tracker		
3	28/07/26	Different types of Network Cables		
4	1/08/26	Setup & Configure LAN using a switch and Ethernet Cables		
5	1/08/26	Experiment on packet Capture to wireshark		
6	13/8/26	Error Correction and detection using hamming Code		

INDEX

Name : Vijay Antony L ..... Std. : ..... Sec. : .....

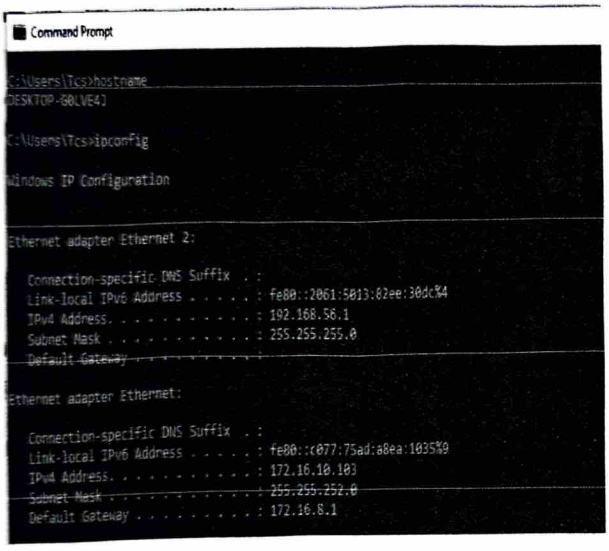
Roll No. : 24181A0431 ..... Sub : Computer Network ....

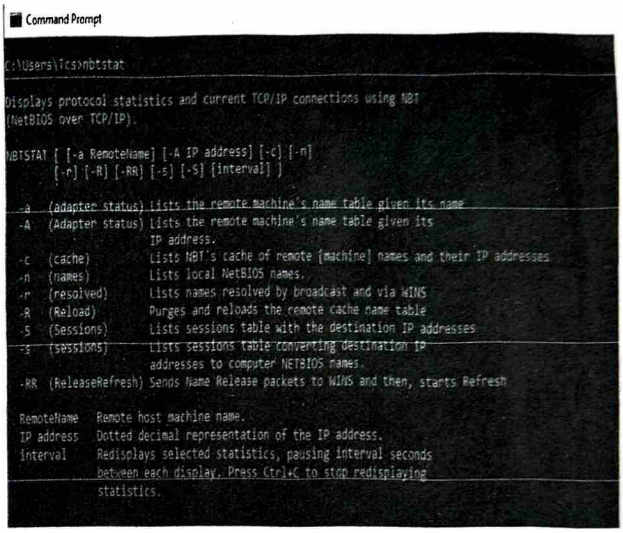
S.No.	Date	Title	Page No.	Teacher's Sign / Remarks
7.		Flow Control at ~~detection~~ Data Link Layer using Sliding window		
8.		8(a) Nmap Live host & Discovery using ARP, ICMP, TCP, UDP 8(b) Configuration of wireless LAN using Cisco Packet Tracer		
9		Implementation of Subnetting in Cisco Packet Tracer Simulation		
10.		a) Internetworking with Router b) Wireless Internetwork		
11		a) Static Routing b) RIP Routing		
12		a) Echo client - Server b) Chat client - Server		
13		Implement your own ping program		
14		Packet Sniffing using Raw Socket		
15		Analysis of various typs of Server		Completed

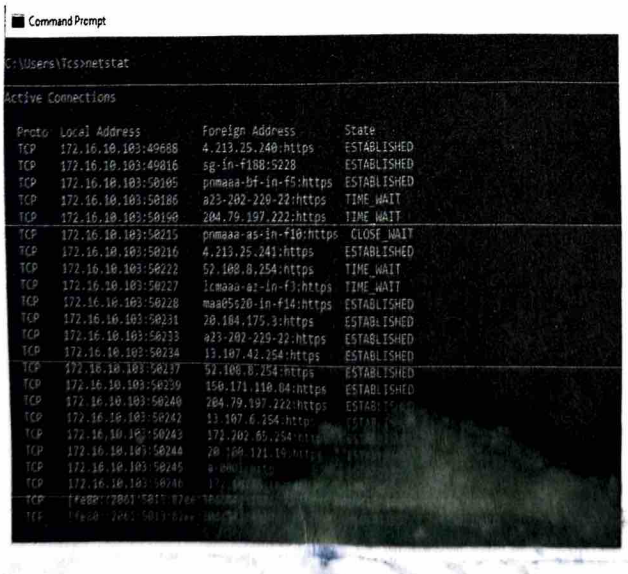
25/07/26

Exp:1 Study of Various Network commands used in Linux and Windows

1) hostname
2) nbstat
3) netstat
4) ipconfig



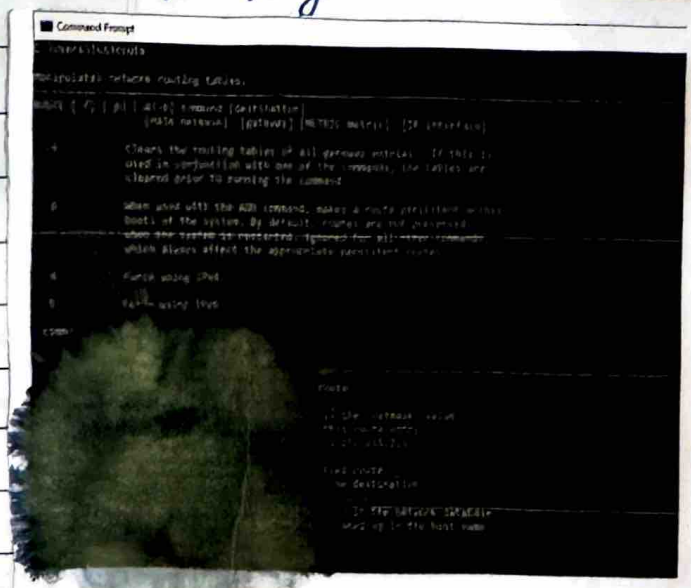


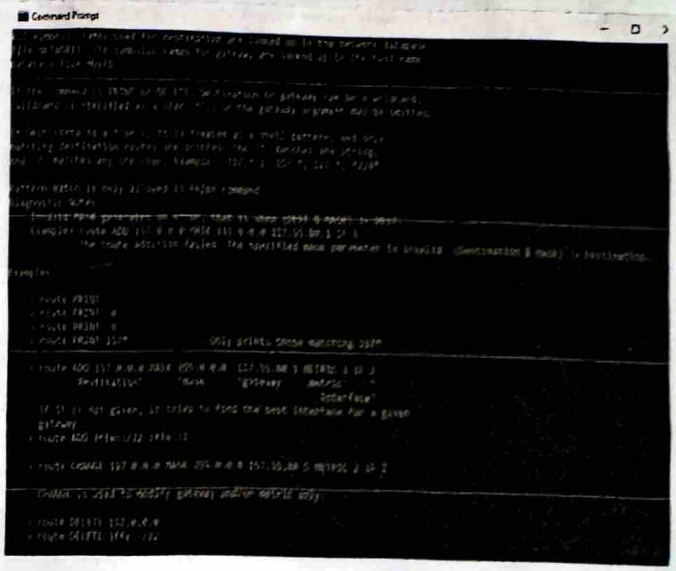


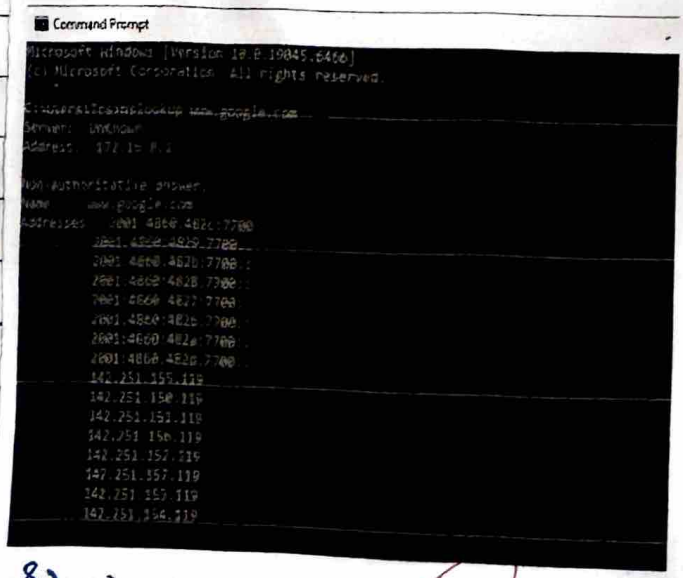
5) route

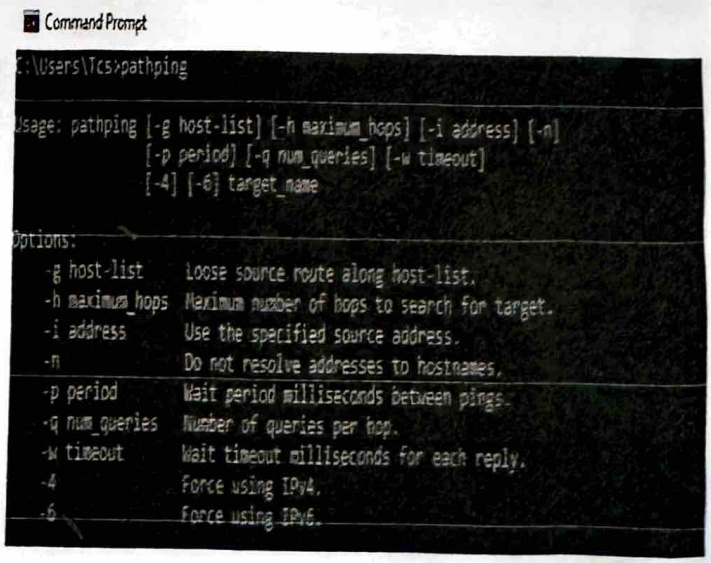
6) lookup

7) pathping

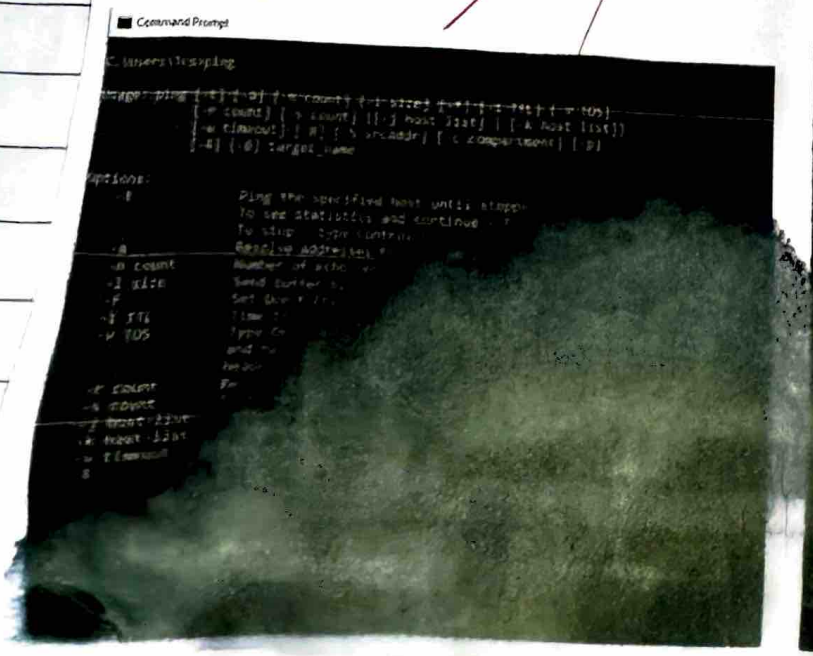




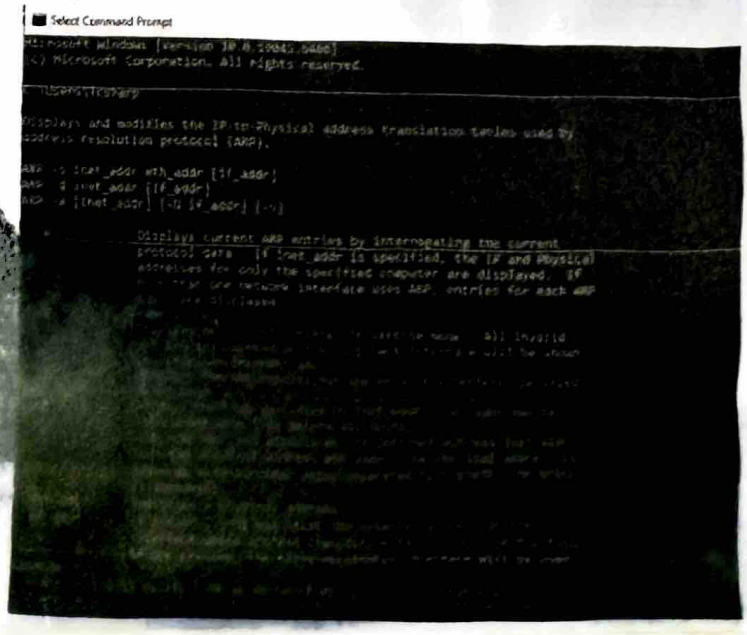




8) ping



9) arp



Linux:

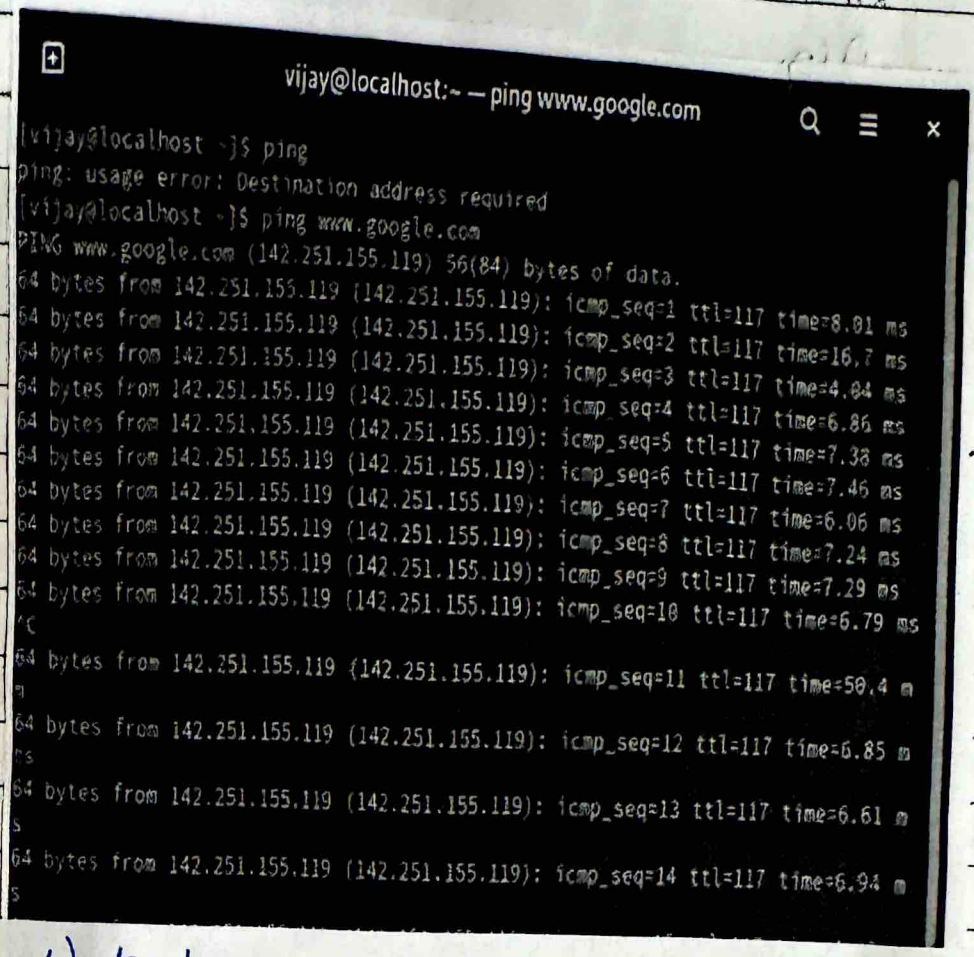
1) ip

```
[vijay@localhost ~]$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ##  {errtoos | kbocos | tcoooe'}
where  OBJECT := { address | addrlabel | route | link | link | neigh | ntoh | ntole|
                   netconf | netwerkk | mrule | | ntte | mcfilter | nrulte |
                   tpfilter | tc|tident | tdata|n | tuntap | maddress | mroute |
                   nefna | nexthop | oee] ntns | netns | l2tp | l3tp | iptnl |
                   flow | token |id | stats | sysctl | help}
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[eatalis] | -f[amily] |
                    -o[neline] <eluan`| -iec | -j[son] | -p[retty] |
                    -j[olo] | -{[oao | -ɔc$ | -ɓ[co] | -j[son | -ɓ[co] |
                    -4 | -6 | -M | -B | -0 |
                    -l[ocks] | -n[umerics]dat[ɨɑ][é-{olor] | -Elicei|[fitntine]

                    -o[neline] = { -t[imestamp] | -ts[hort] | -batch|[F]ilename}
                                 }
                    -r[esolve] -s[ize] | -n[ames name | -N[umeric] | -a[ll] |
                    -c[olor]}
[vijay@localhost ~]$
```

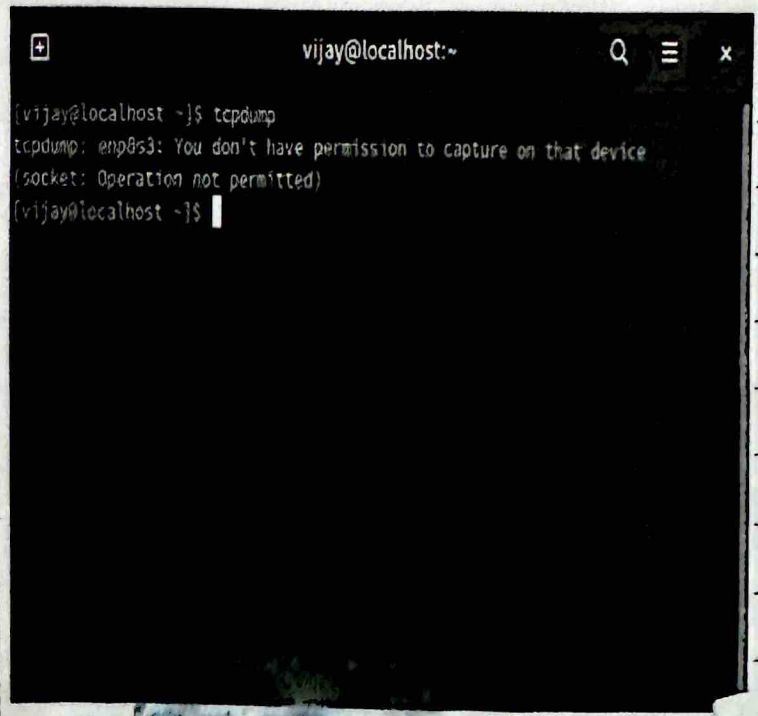
) ipconfig

```
[vijay@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:feff:4b3e  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:ff:4b:3e  txqueuelen 1000  (Ethernet)
        RX packets 23  bytes 2983 (2.8 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 35  bytes 3665 (3.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 26  bytes 2407 (2.3 KiB)
        RX errors 0  dropped 0  overruns 0  fram
        TX packets 28  bytes 2407 (2.3 KiB
        TX errors 0  dropped 0 overruns

[vijay@localhost ~]$
```

3) ping -?



4) tcpdump



5) mtr -?

Result:

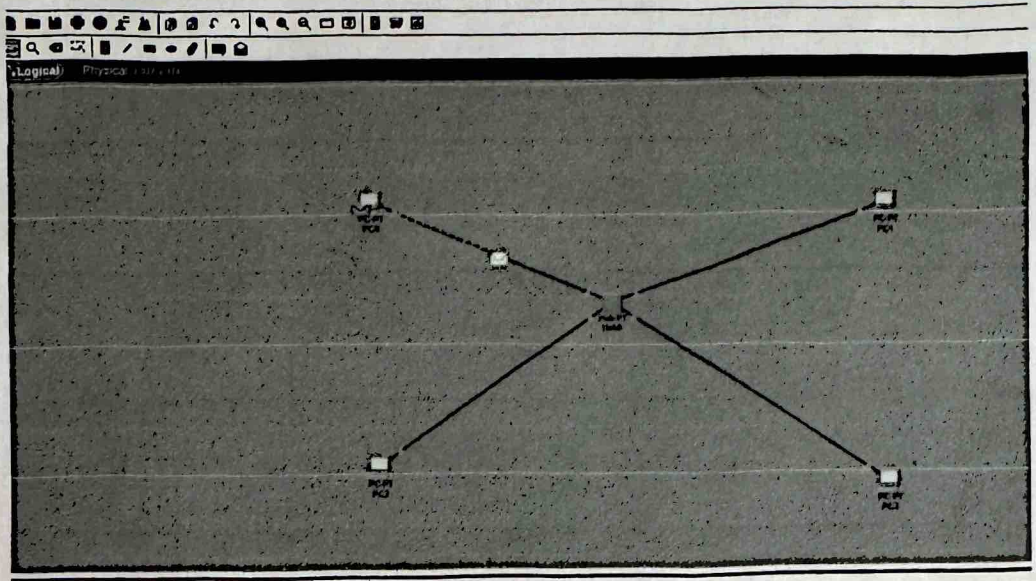
Thus the network commands used in windows & Linux are implemented successfully.

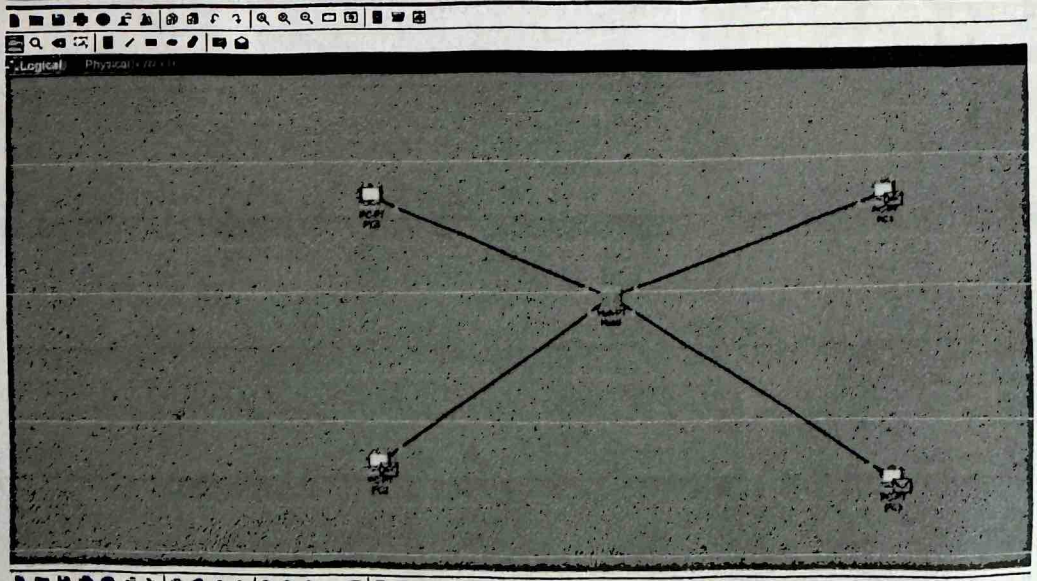
DATE :
PAGE :

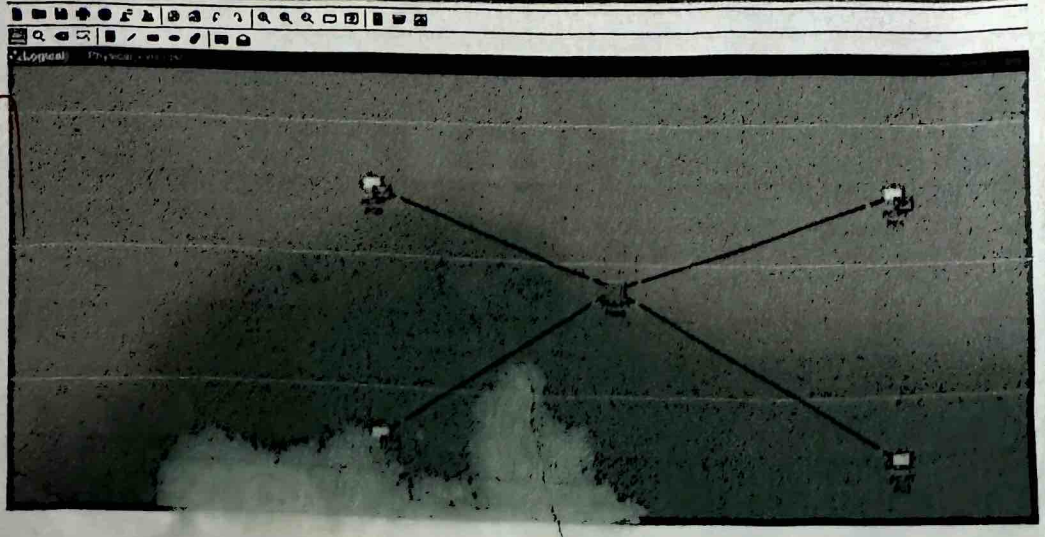
1/08/26

Exp:03 Experiment on CISCO packet Tracker

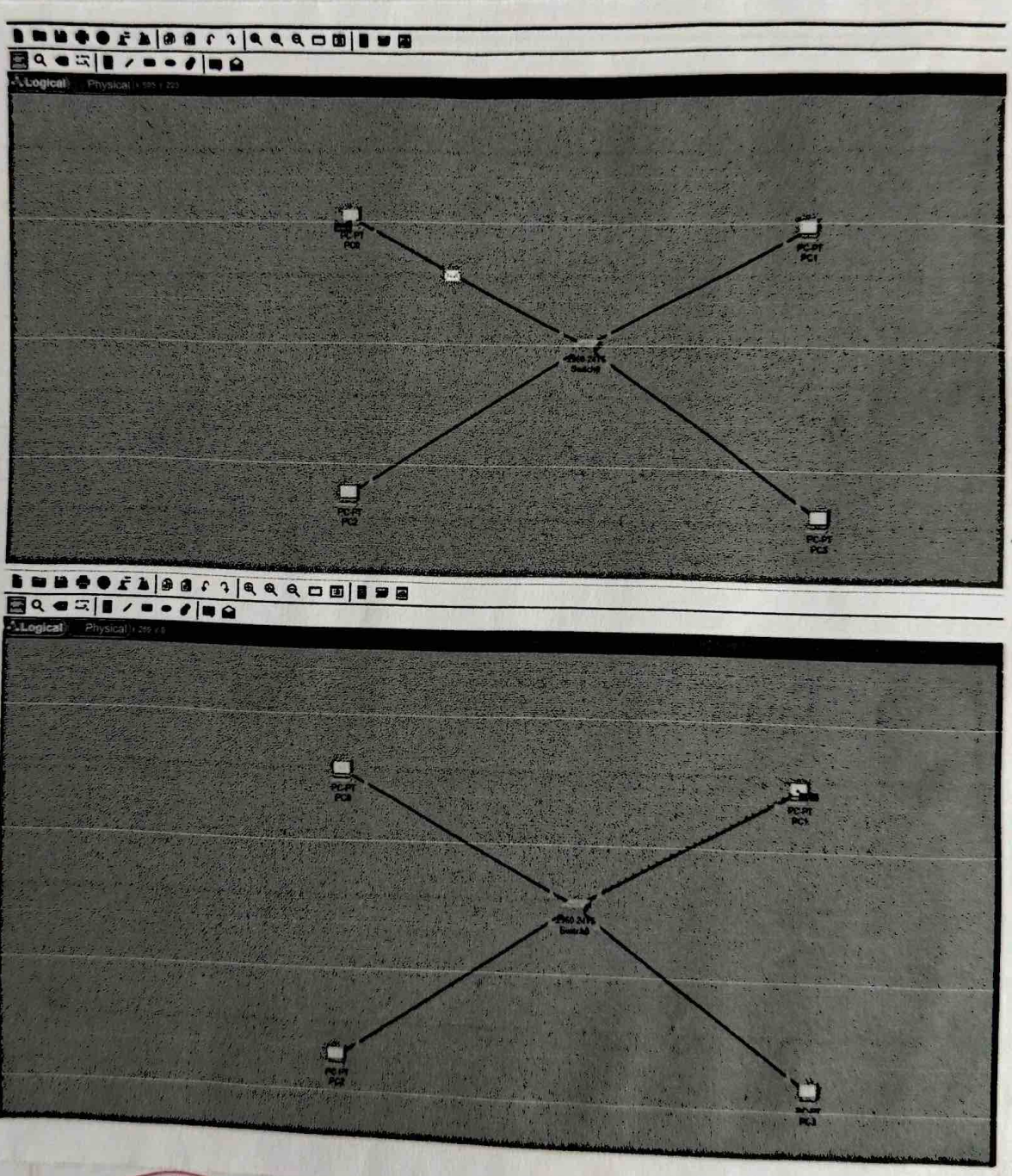
2a) Hub:

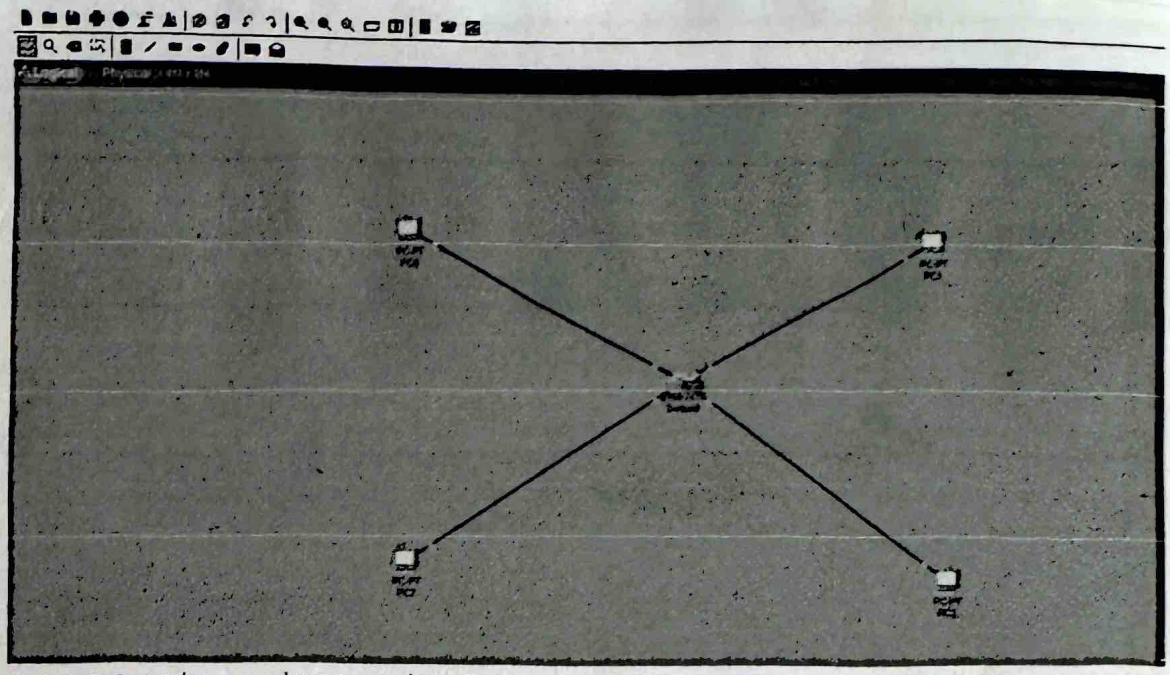


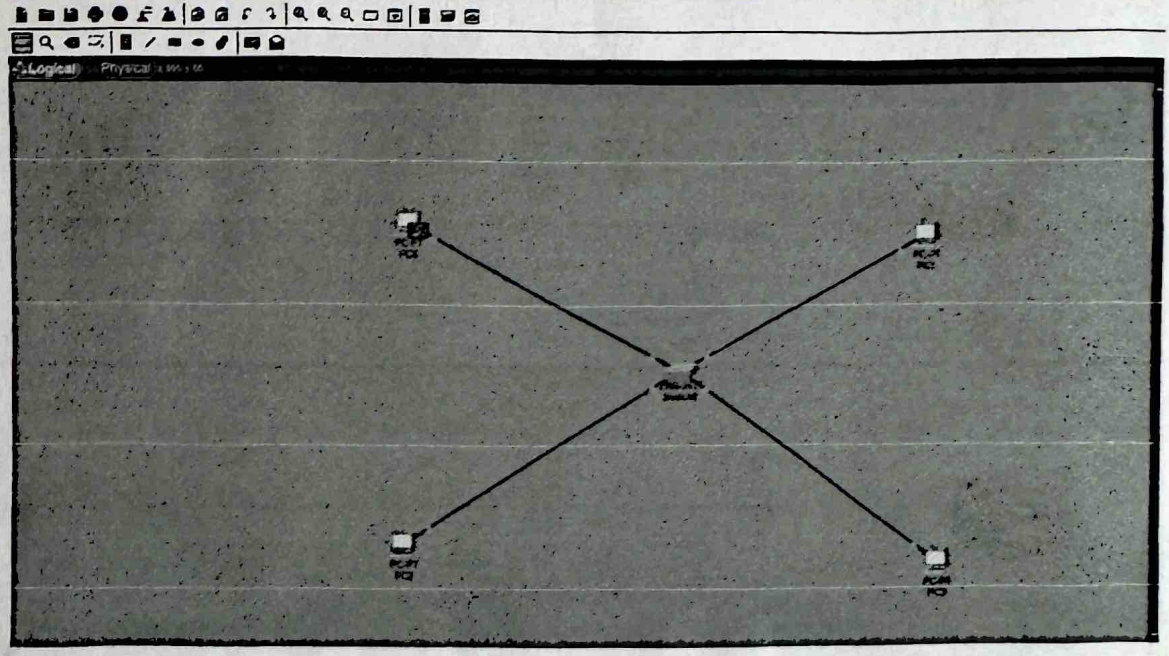




3b) Switch:


Logical
Physical
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3
Logical
Physical
PC-PT PC0
PC-PT PC1
PC-PT PC2
PC-PT PC3





Procedure:

1) From the network component box, click and drag and drop
   a) 4 Generic PC and One Hub
   b) 4 Generic PC and One switch

2) Click on connections

3) click the PC connected to hub and

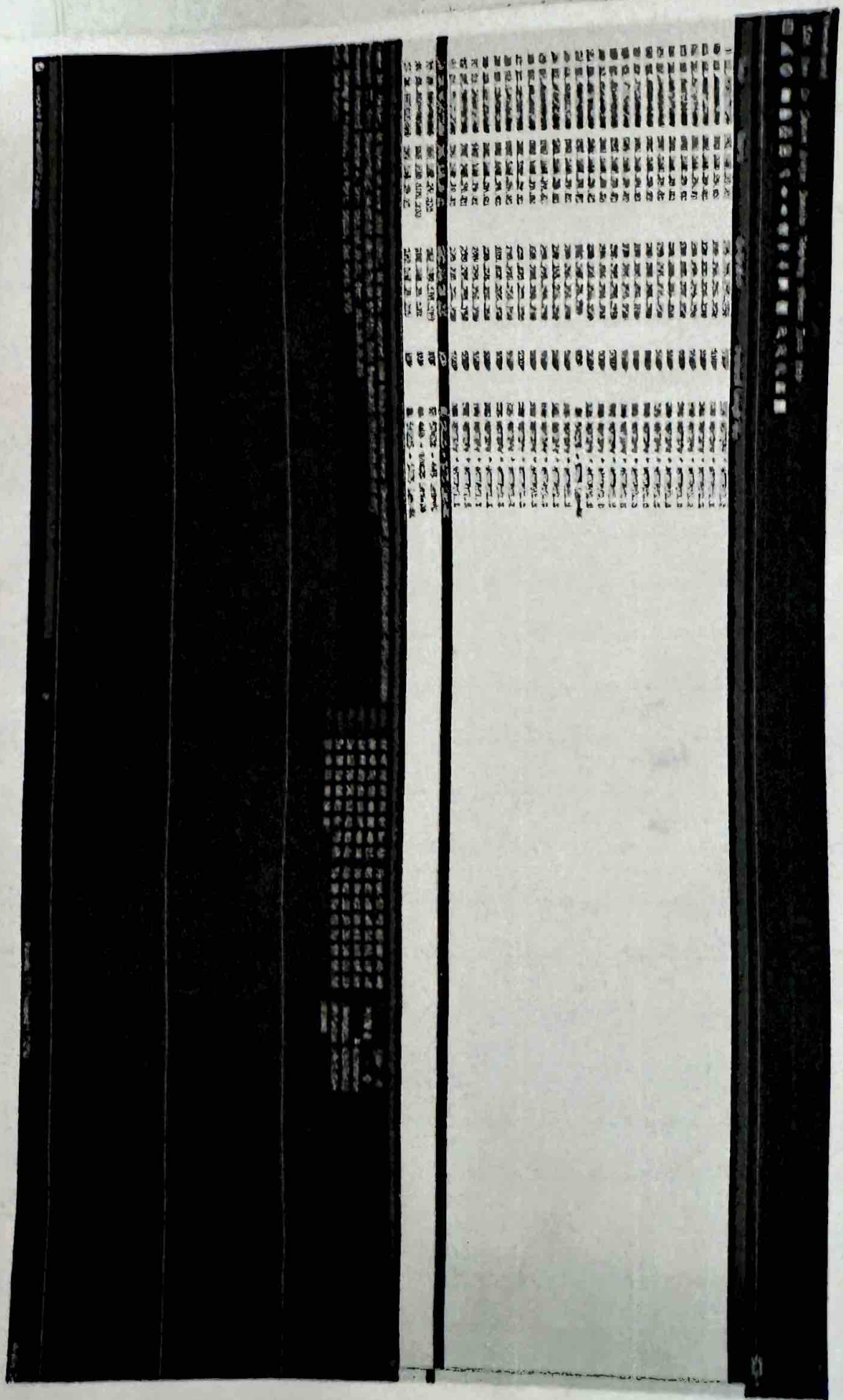
change the IP configuration.

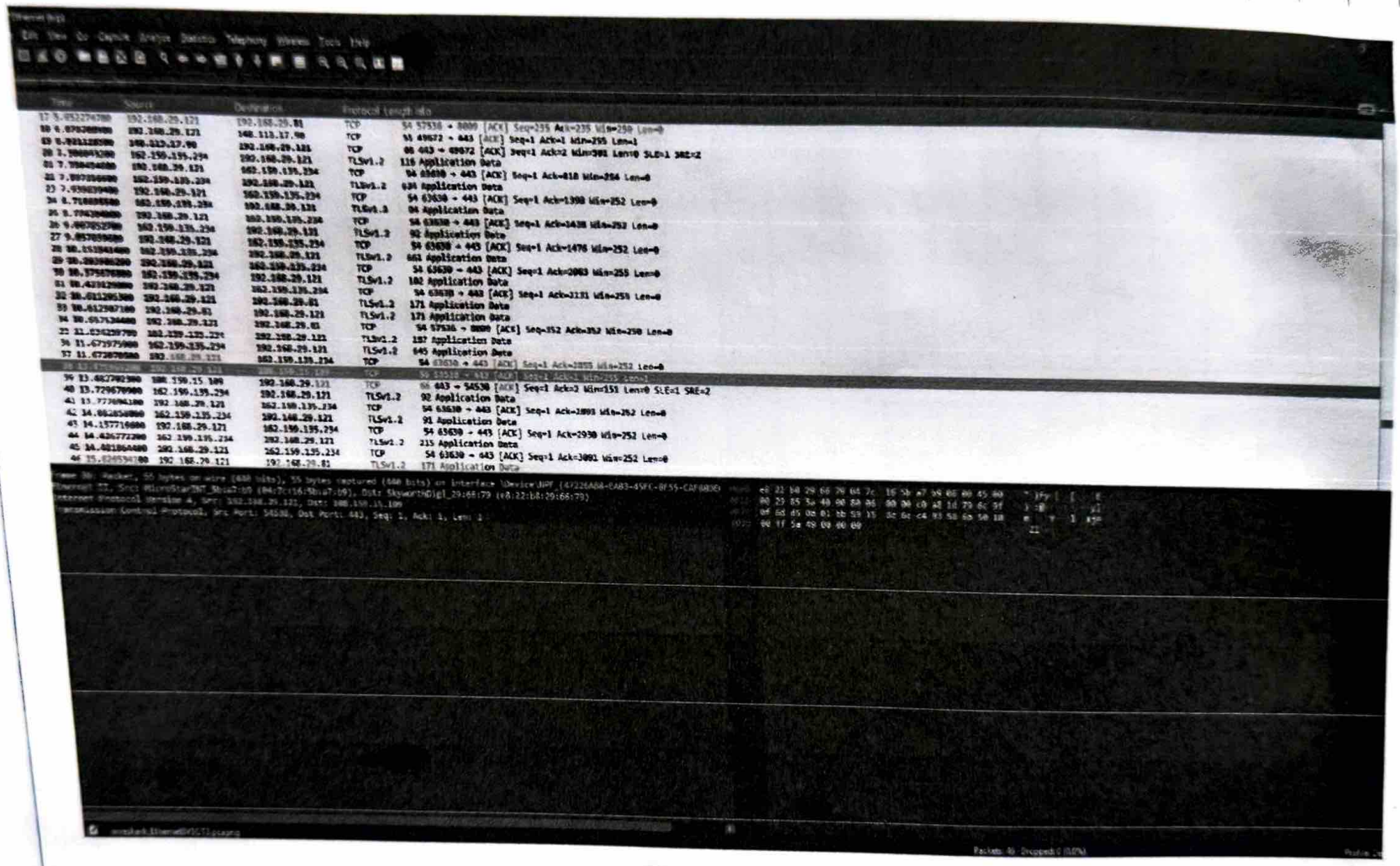
4) Observe the flow of PDU from source PC to destination PC.

Result :-

Thus to design a simple network with multiple nodes and connected via networking devices available in library is implemented successfully.

Aim: - To learn Capture, views, packets in wireshark tool





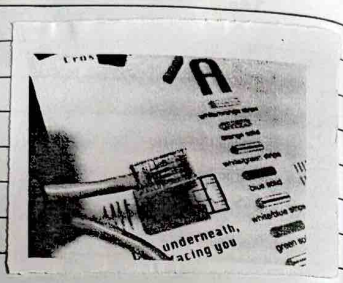
Procedure:

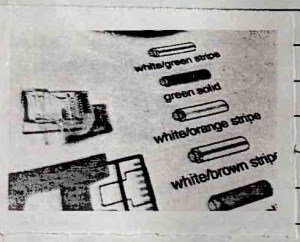
* Select Local Area Connection in wireshark
* go to capture → option
* Select stop capture automatically after 100 packets.
* Then click start capture
* Search TCP/UDP in search bar
* click statistics → Flow Graph
* Save the packets

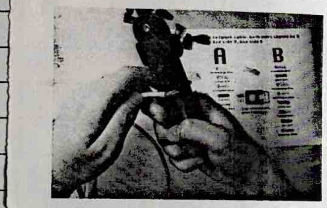
Result:

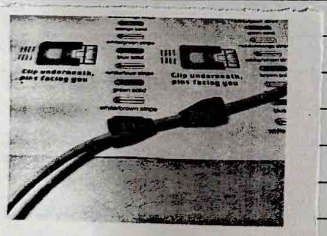
Thus the set up and configure a LAN using a switch and Ethernet cable as

Thus, the experiment on packet capture tool wireshark has been implemented and verified.









Ex no: 2 Study of Network Cables

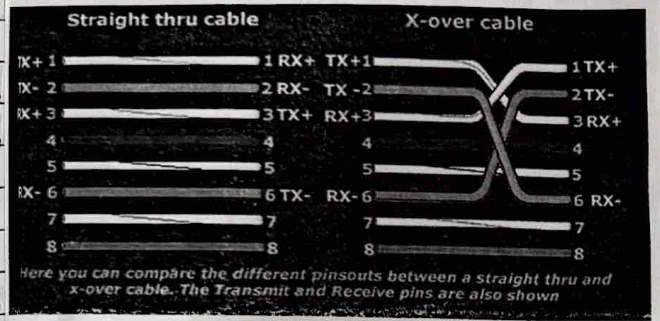
Aim:

Study of different types of Network cables

- Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) cable
2. Shielded Twisted Pair (STP) cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable type	Category	Maximum Data Transmission	Advantages / Disadv	Application
UTP	Category 3	10 bps	Adv: * Cheaper in cost * Easy to install Dis: * More prone to EMI and noise	10 Base - T Ethernet
	Category 5	upto 100 Mbps		Fast Ethernet
	~~Category 5e~~	~~1 Gbps~~		Gigabit Ethernet
STP	Category 6, 6a	10 Gbps	Adv: * Shielded * Faster than UTP Dis: Expensive Greater install effort	Gigabit 10 G Ethernet (55 m)
SSTP	Category 7	10 Gbps		Gigabit Ethernet, 10G Ethernet (600m)



			Adv:	
Coaxial	RG-6 RG-59 RG-11	10-100Mbps	* High bandwidth * Immune to interference Dis: * Limited distance * Cost	. Speed of signal is 500M Television Network
Fibre Optics	Singlemode Multimode	100Gbps	Adv: * High speed * High bandwidth Dis: * Expensive * Requires Skilled installers	*Maximum distance of fibre is 100 meters

Procedure:

* ~~Step 1:~~
* To start construction of the device, begin by threading shields onto the cable

Step 2:
* Strip approximately 1.5cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3:
* You will need to untangle the wires: there should be four "twisted pairs" Referencing back to the sheet, arrange them from

top to bottom. One end should be in arrangement A and other in B.

Step 4:
Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into RJ45 plug without messing up the order.

Step 5:
Push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding

Step 6:
After the wires are securely siting inside the plug, insert it into the crimping tool and push down

Step 7:
Repeat the for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straight through cable)

Result:
Thus, The different type of network cable has been studied and implemented successfully

Exno : 4 Setup and Configure a LAN Local Area Network using a switch and Ethernet Cables

Aim:

To setup and configure a LAN using a switch and Ethernet Cables.

What is LAN?

A local Area Network refers to a network that connects device within a limited area, such as an office building, school or home. It enables users to share resource, including data, printers and internet access.

How to setup a LAN :-

Step 1:-

Plan and Design an approprite network topology taking into account network requirments and equipment location.

Step 2:

You can take 4 Computer, Switch with 8, 16 or 24 ports which is sufficient for networks of these sizes and 4 Ethernet Cables.

Step 3:

Connect your Computer to the network switch via an Ethernet cable, which is as simple as plugging one end of the Ethernet Cable into your computer and other

Step 4: ~~Assign IP address~~

* Assing IP address to your Computer

eg: PC1 - IP address : 10.1.1.1, subnet mask 255.0.0.0

Step 5:

* Configure a network switch

* Assign IP address as : 10.1.1.5, Subnet mask 255.0.0.0

Step 6:

Check the connectivity between switch and other machine by using ping command prompt of the device.

Step 7:

Select a folder → go to properties → click Sharing tab → share it with everyone on the same LAN.

Step 8:
Try to access the shared folder from others computer of the network

Result:
Thus, the setup and Configure a LAN using a switch and Ethernet has been implemented and executed successfully.

Ex no: 6 Error Detection & Correction using Hamming Code

Aim :-

To write a program to implement error detection & Correction using hamming code.

Algorithm :-

* Read the binary data bit.
* Count the no. of data bits m.
* Find the number of parity bits using $2^r \geq m + r + 1$
* Create positions for Hamming Code.
* Place parity bits at positions 1, 2, 3, 4, 8
* Calculate each parity bits ~~at position~~ using XOR
* Display the Hamming Code.
* Read Code and Check the parity bits
* If error position is not zero flip the bit to correct the error.

Code:

```
def calculate_parity_bits(data):
    m = len(data)
    r = 0
    while (2**r) < (m+r+1):
        r += 1
    return r

def generate_hamming_code(data):
    m = len(data)
    r = calculate_parity_bits(data)
    n = m+r
    hamming = ['0'] * (n+1)
    j = 0
    for i in range(1, n+1):
        if i & (i-1):
            hamming[i] = data[j]
            j += 1
    for p in range(r):
        position = 2**p
        parity = 0
        for i in range(1, n+1):
            if i & position:
                parity ^= int(hamming[i])
        hamming[position] = str(parity)
    return ''.join(hamming[1:])

def detect_and_correct(recived):
    n = len(recived)
    r = 0
    while (2**r) < (n+1):
        r += 1
    hamming = ['0'] + list(recived)
    error_position = 0
    if error_position != 0:
        print("Error position",
              error_position)
        if hamming[error_position]
    else:
        hamming[error_position] =
    return "".join(hamming[1:])
print("hamming code")
data = input("Enter the binary data) &
if not all(bit in '01' for bit in data):
    print("please enter only 0 and 1")

else:
    hamming_code = generate_hamming
        _code(data)
    print("Hamming code", hamming
        _code)
    if len(recived) != len(hamming
        code) or not all(
        bit in '01' bit in recived):
    else:
        corrected_code = detect_and_
        correct(recive
```

Output:

```
Enter the binary data: 1011
original Data : 1011
Hamming code : 0110011
Enter the recived Hamming code: 0110111
Enter position : 5
Error detected at position 5
Error corrected successfully!
Recived code : 0110111
Corrected code : 0110011
```

Result:

The hamming code problem was successfully implemented in Python

Exp No: 7

Date
Page

Flow Control at Data Link Layer using Sliding Window Protocol

Aim:

To write a program to implement flow control at the Data Linker layer using the sliding window protocol and to simulate the flow of frames from one node to another.

Algorithm:

1. Start the program
2. Read the total number of lines to be sent (N) the sender's window size (W) and a frame no. to simulate a lost destiny transmission.
3. Initialize base = 0 and next seq = 0
4. Repeat while base < N: determine the sender ~~Deter~~ window from base to base+W-1
5. Send all frames within the current window and increment next sequence after sending each frame.
6. If the specified frame is lost:
   - Reciver deletes the missing frame
   - Reciver discards the out-of-order frame
   - Sender Experience a timeout
7. Repeat steps 4-7 until all frames are succesfully transmitted acknowledg
8. Display all frames
9. Stop the program.

Code:

```
import java.util.scanner;
public class Sliding Window {
    static void simulate (int total frames,
        int window size, Integer lost Frame){
            int base=0;
            int next seq =0;
        while (base < total Frames) {
            int window End = math. min (
                base + window Size - 1)
            System.out.println("Sender Window:[
                frame " + base + " frame " + window
                End + "]");
            while (next Seq < base + window Size
                next Seq < total frames ) {
                syten. out. println ("Node A →
                    Sending Frame " + next Seq);
                if (lost Frame != null && next seq ==
                    lost Frame ) {
                    System.out.println ("Frame " +
                    next seq + "lost in transit !) ");
                    } next seq;
                }
            if (lost Frame != null && base <= lost Fram
                && lost frame < next Seq ){
                System.out.println ("Node B → Frame " +
                    lost frame + "missing ");
                System.out.println (
                    " Node A → Time out ! Resending
                    window starting from frame " +
                    lost frame " . " );
                next Seq = lost Frame ;
```

```
lostFrame == null;
continue;
for (int j = base; j < next Seq; j++) {
    System.out.println(
        "Node B -> Recived Frame " + j + ", Sending
            Ack" + j);
    }
    base = next seq;
  }
}
public static void main (String[] args) {
    Scanner sc = new Scanner (System.in);
  System.out.println(
        "Enter total number of frames to send");
    int totalFrames = sc.nextInt();
      System.out.println(
        "Enter frames numbers to stimulate
            as lost (n "-1)
        " (0 to " + (totalframes - 1) + ";
                enter -1 for no loss):"
        );
  Select
    lostFrame = input lostFrame;
  }
  simulate (total frame, windowSize,
            lostFrame);
      sc.close();
  }
}
```

Output:

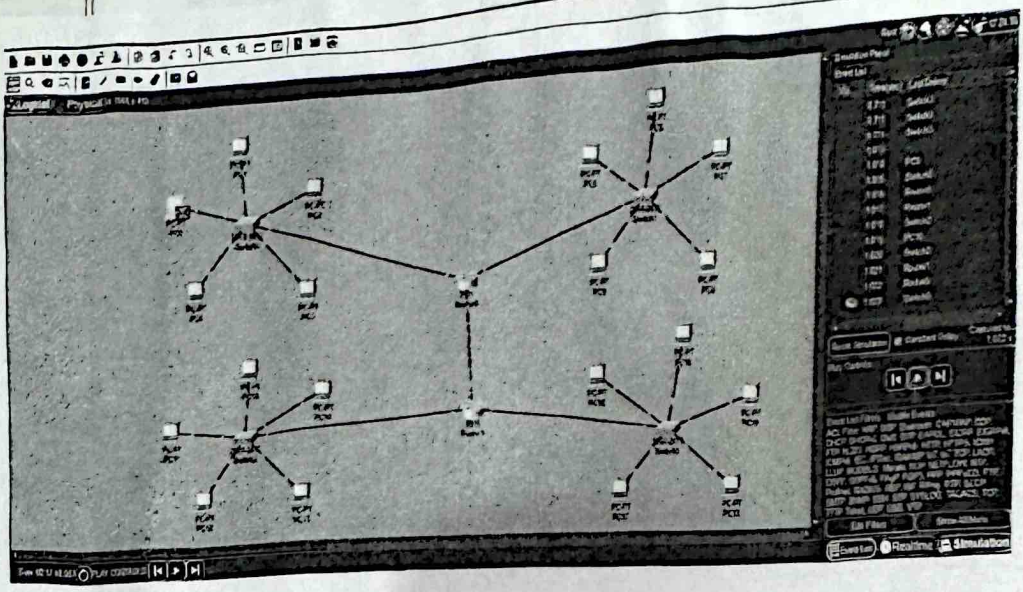
Enter total number of frames to send: 3
Enter Senders window size = 2
Enter frame number to simulate as lost
(0 to 2, enter -1 for no loss): 1

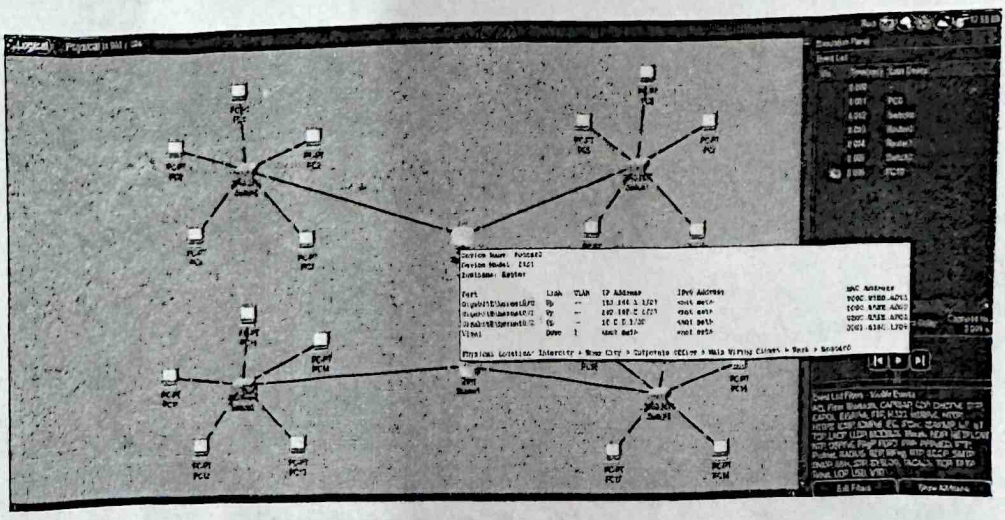
Sender Window [Frame 0 - Frame 1]
Node A → Sending Frame 0
Node B → Sending Frame 1
(Frame 1 lost in transit)
Node B → Frame 1 missing
Node A → Time out! Resending
frame frame 1

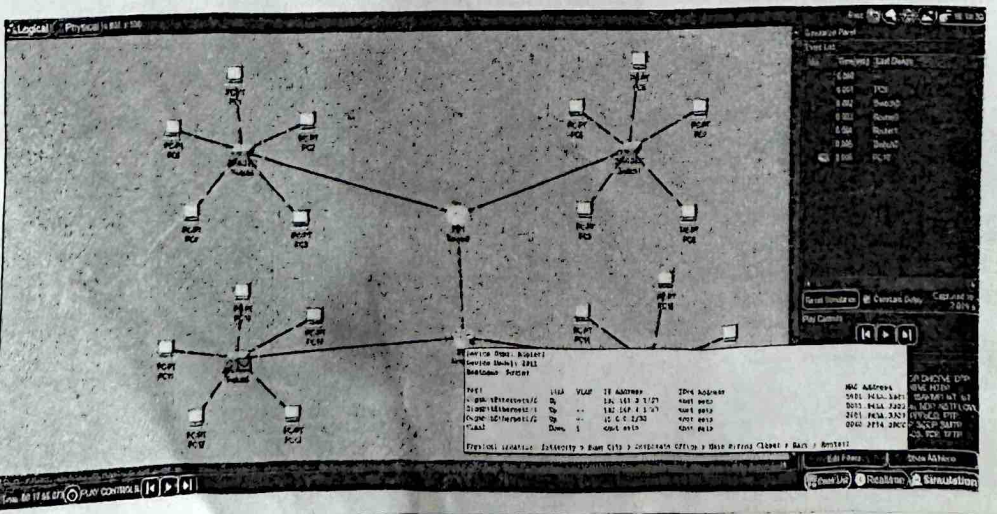
Sender window: [Frame 1 - Frame 2]
Node A → Sending Frame 1
Node A → Sending Frame 2
Node B → Recieved Frame 1, Sending
Ack 1
All 3 frames successfully transmitted
from A to Node B

Result:

Thus the program to implement flow control of data link layer using the sliding window protocol was implemented successfully







Ex no: 9

Date : 24/09/26

Implementation of Subnetting in CISCO packet Tracker simulator

Aim:

To implement subnetting in cisco packet Tracer simulator by creating multiple subnets, assigning static IP addresses, configuring routers and establishing communication between different subnets.

Procedure:

1) Open Cisco Packet Tracer and create a new network topology.

2) Add 2 routers, 4 switchs and 20 pcs and connect them using appropriate cables.

3) Consider the network 192.168.1.0/24 and divide it into smaller subnets using /27 subnet mask.

   255.255.255.244

4) Configure Router 0:

   G0/0 → 192.168.1.127

   G0/1 → 192.168.2.127

   G0/2 → 10.0.0.1/30

5) Configure Router 1:

   G0/0 → 192.168.3.1/27

   G0/1 → 192.168.4.1/27

   G0/2 → 10.0.0.2/30

6) Enable all router interfaces using the no shutdown command.

7) Assign Static IP addresses to the PCs:
LAN 1: 192.168.1.11 - 192.168.1.15
LAN 2: 192.168.2.11 - 192.168.2.15
LAN 3: 192.168.3.11 - 192.168.3.15
LAN 4: 192.168.4.11 - 192.168.4.15

8) Assign the subnet mask 255.255.255.224 and the apporiate gateway to each PC.

9) Configure static router between Router 0 and Router 1:

Router 0:
ip route 192.168.3.0 255.255.255.244 10.0.0.2
ip route 192.168.4.0 255.255.255.244 10.0.0.2

Router 1:
ip router 192.168.1.0 255.255.255.244 10.0.0.1
ip router 192.168.2.0 255.255.255.244 10.0.0.1

10) Test communication between PCs in different Subnets using the ping command.

11) Switch the simulation mode and sent a simple PDU between PCs from diffent subnets.

Date
Page

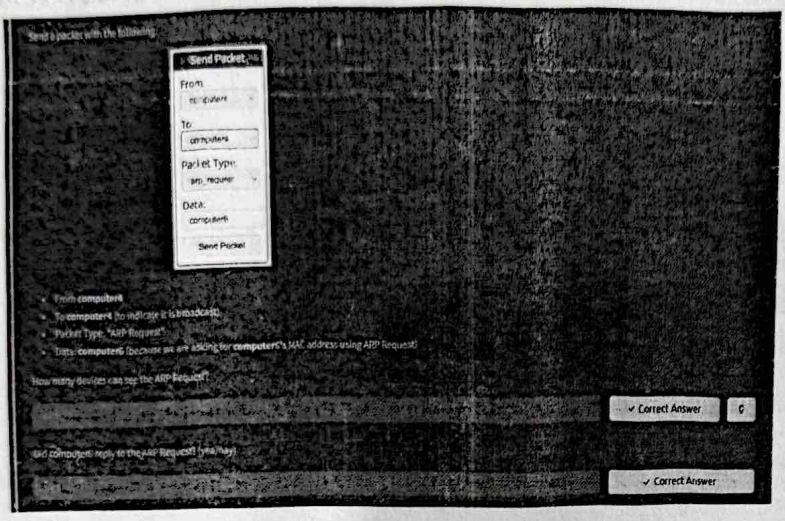
2) Observe the packet travelling through the switches and routers ~~PC's from~~ different ~~subnets~~ and verify successful communication.

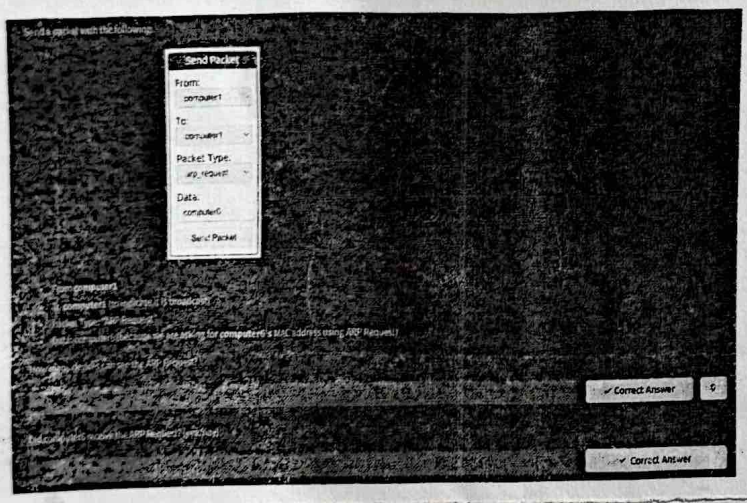
Result

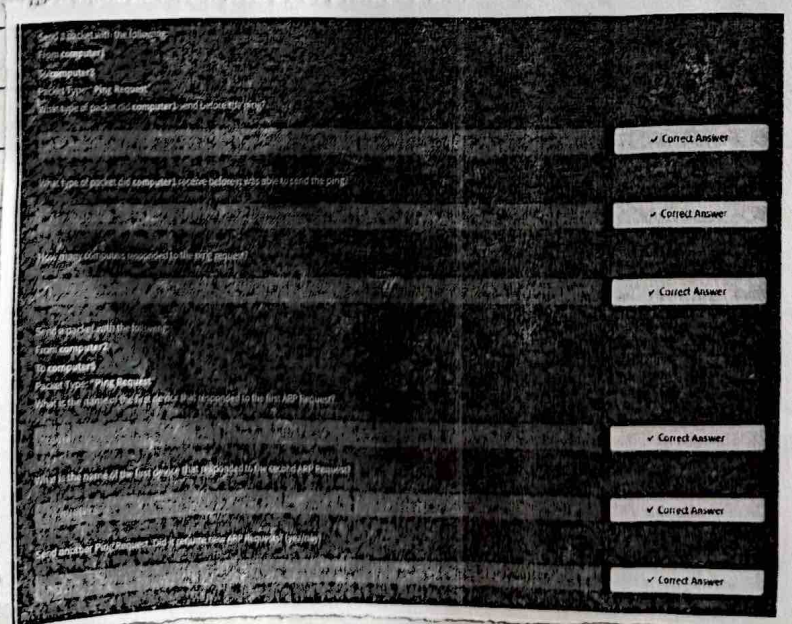
Thus, the subnetting was successfully implemented in cisco packet tracker.

Exp no: 8(a)

Date
Page







...ining the CONNECTION_IP/24?

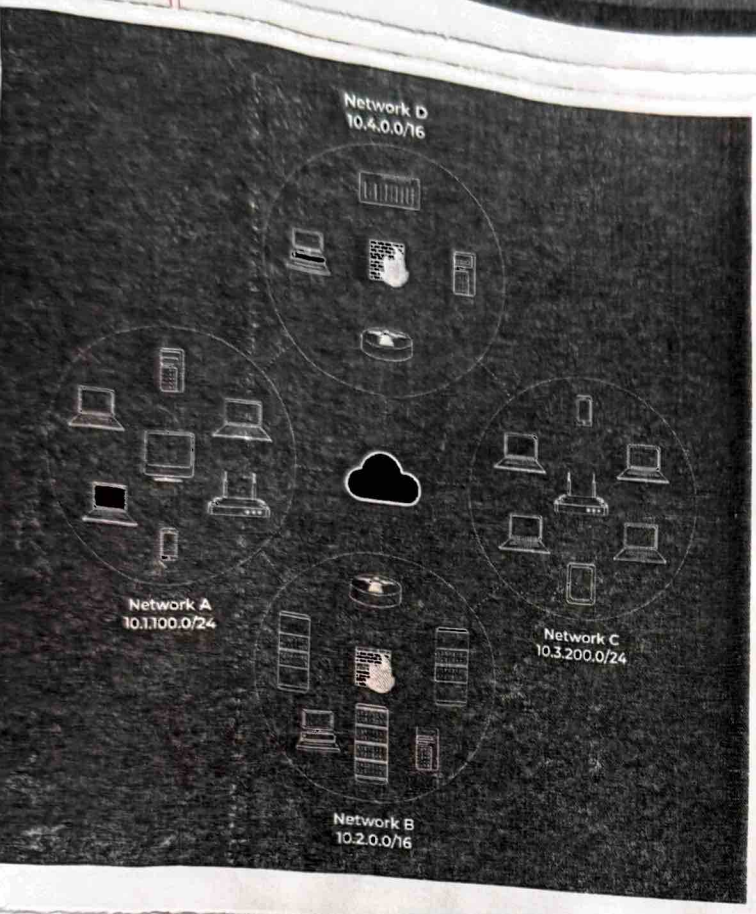
✓ Correct Answer

What is the first IP address Nmap would scan if you provided 10.10.12.13/29 as your target?

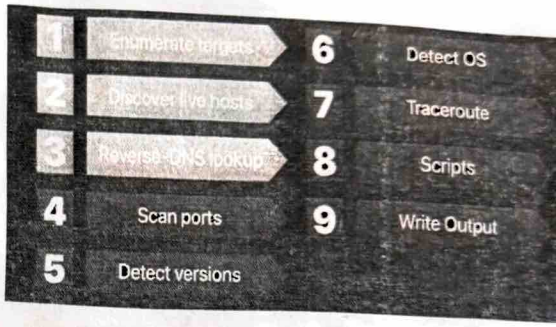
✓ Correct Answer

How many IP addresses will Nmap scan if you provide the following range: 10.10.0.255-101.125?

✓ Correct Answer



	ISO/OSI	TCP/IP
7	Application Layer	Application Layer
6	Presentation Layer	
5	Session Layer	
4	Transport Layer	Transport Layer
3	Network Layer	Network Layer
2	Data Link Layer	Link Layer
1	Physical Layer	



Answer the questions below

We want Nmap to issue a reverse DNS lookup for all the possible hosts on a subnet, hoping to get some insights from the names. What option should we add?

✓ Correct Answer

Answer the questions below

Which TCP ping scan does not require a privileged account?

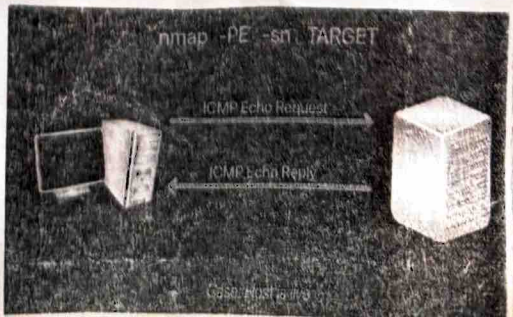
✓ Correct Answer

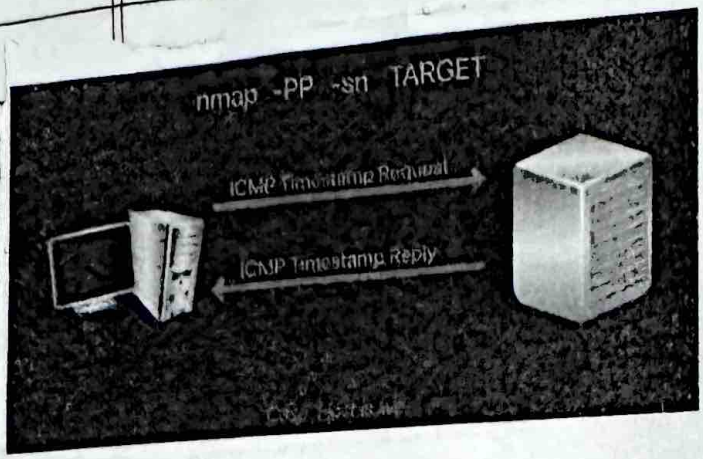
Which TCP ping scan requires a privileged account?

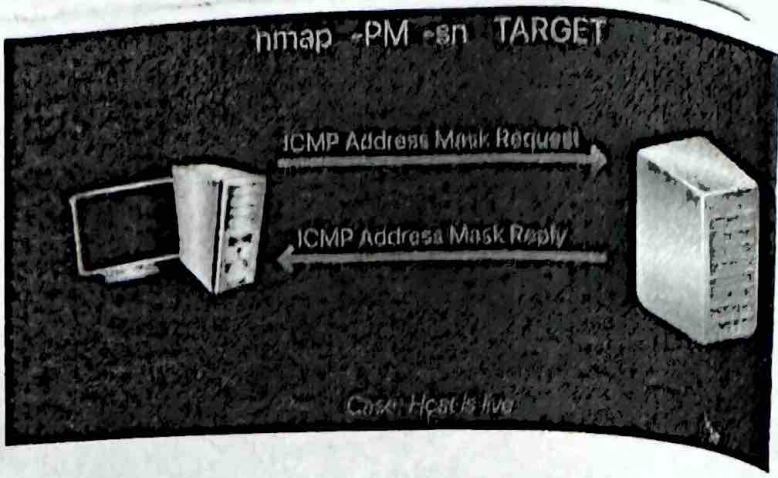
✓ Correct Answer

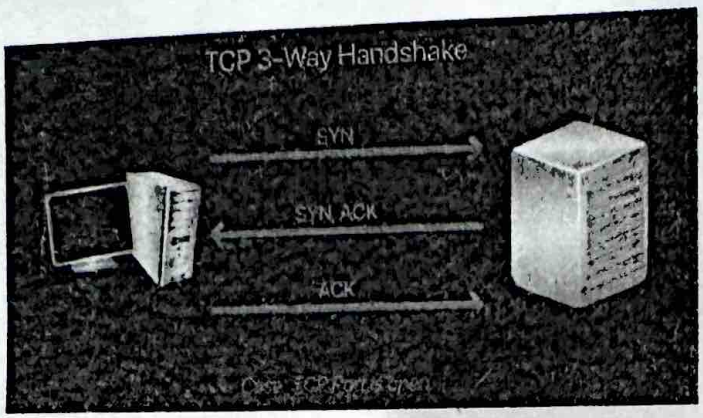
What option do you need to add to Nmap to run a TCP SYN ping scan on the telnet port?

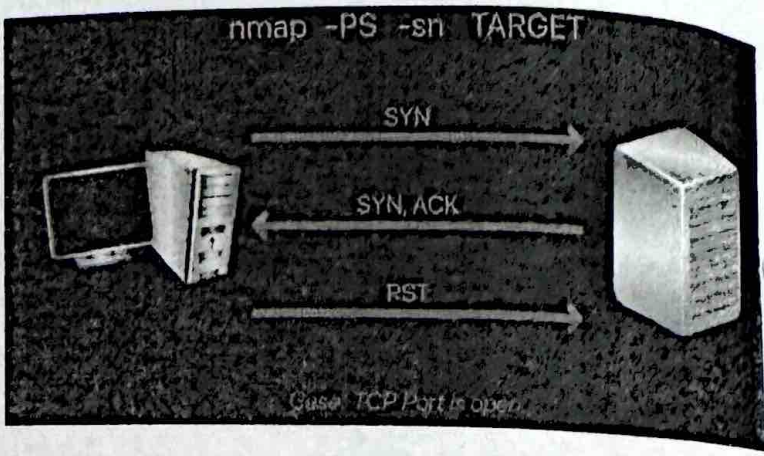
✓ Correct Answer

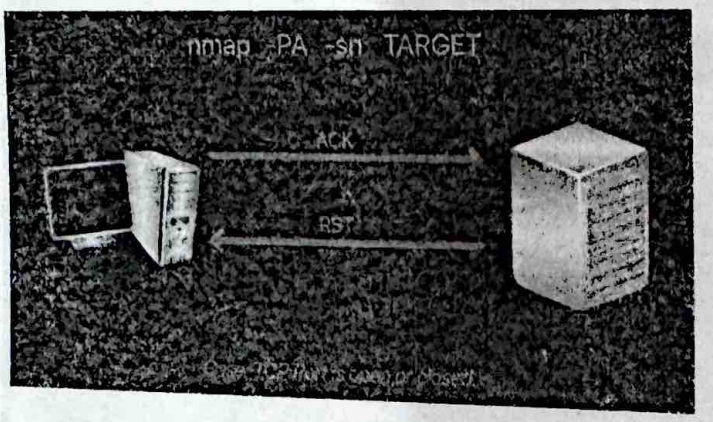


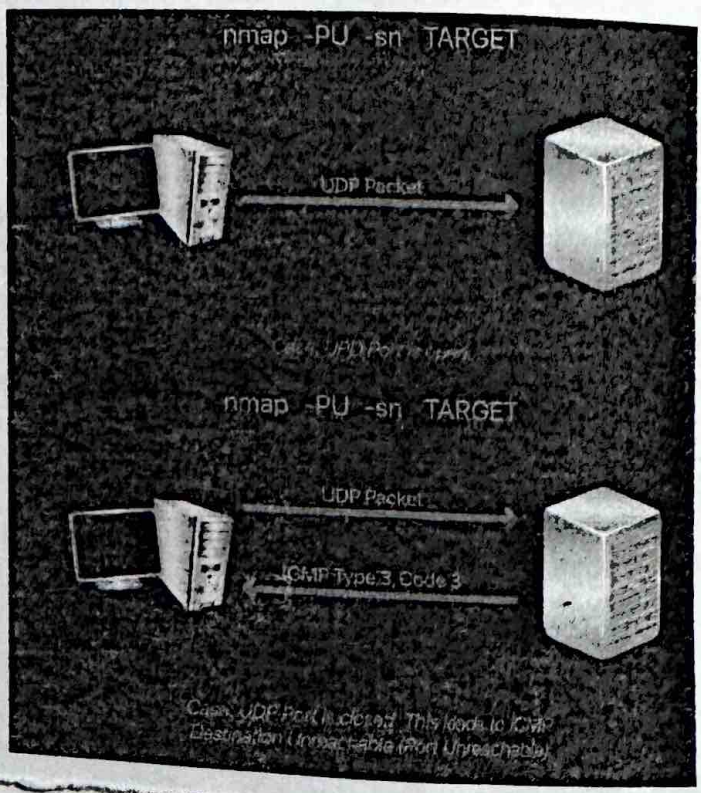

nmap -PP -sn TARGET
ICMP Timestamp Request
ICMP Timestamp Reply

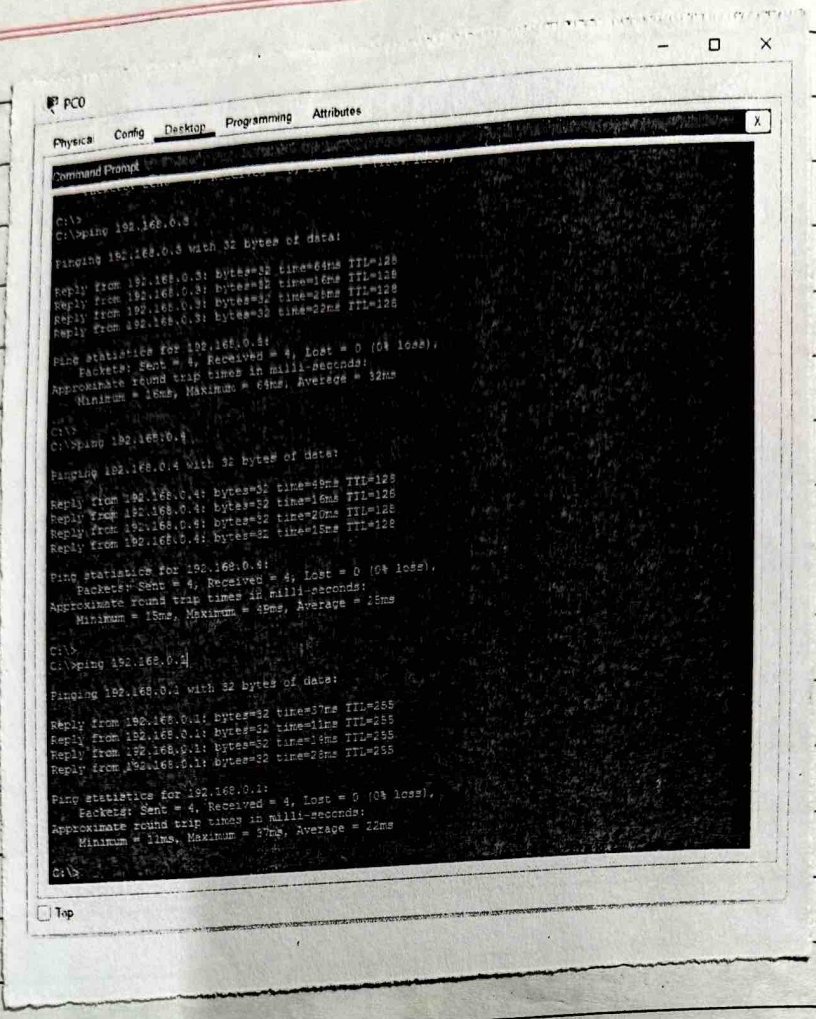

nmap -PM -sn TARGET
ICMP Address Mask Request
ICMP Address Mask Reply

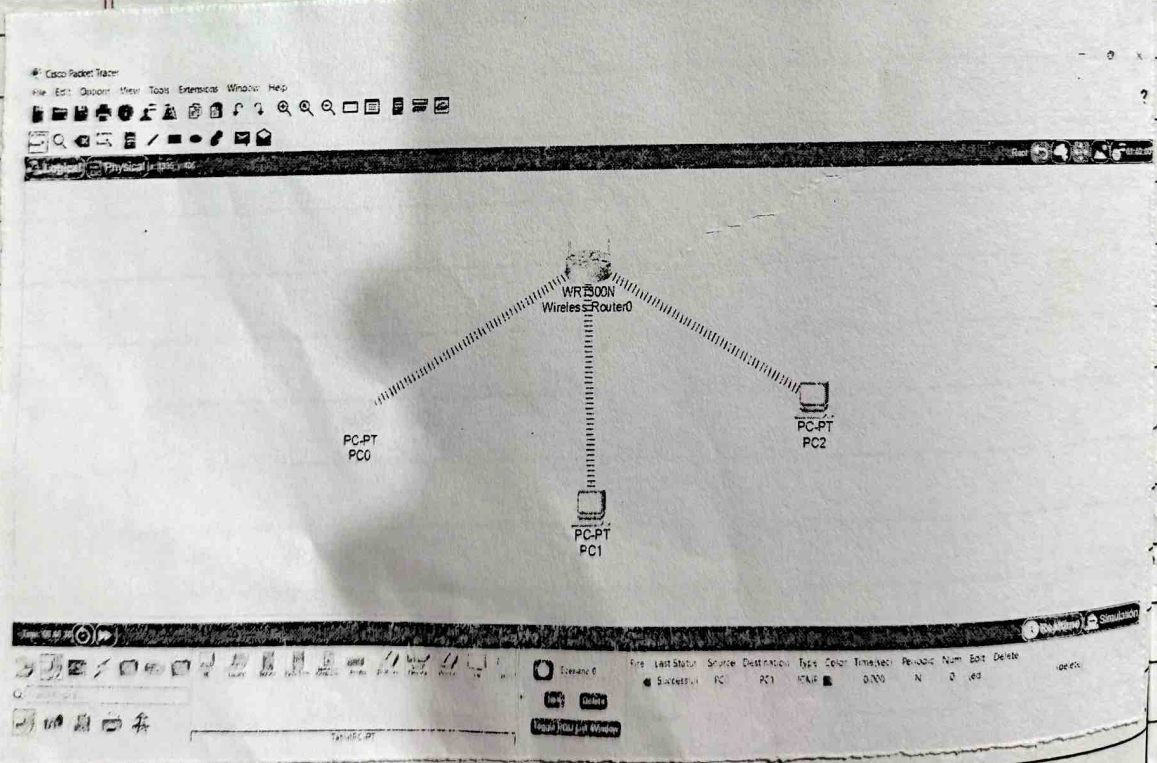

TCP 3-Way Handshake
SYN
SYN, ACK
ACK


nmap -PS -sn TARGET
SYN
SYN, ACK
RST


nmap -PA -sn TARGET
ACK
RST


nmap -PU -sn TARGET
UDP Packet
nmap -PU -sn TARGET
UDP Packet
ICMP Type:3, Code 3


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>
C:\>ping 192.168.0.3
Pinging 192.168.0.3 with 32 bytes of data:
Reply from 192.168.0.3: bytes=32 time=64ms TTL=128
Reply from 192.168.0.3: bytes=32 time=16ms TTL=128
Reply from 192.168.0.3: bytes=32 time=28ms TTL=128
Reply from 192.168.0.3: bytes=32 time=22ms TTL=128
Ping statistics for 192.168.0.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 16ms, Maximum = 64ms, Average = 32ms
C:\>
C:\>ping 192.168.0.4
Pinging 192.168.0.4 with 32 bytes of data:
Reply from 192.168.0.4: bytes=32 time=49ms TTL=128
Reply from 192.168.0.4: bytes=32 time=16ms TTL=128
Reply from 192.168.0.4: bytes=32 time=20ms TTL=128
Reply from 192.168.0.4: bytes=32 time=15ms TTL=128
Ping statistics for 192.168.0.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 15ms, Maximum = 49ms, Average = 25ms
C:\>
C:\>ping 192.168.0.1
Pinging 192.168.0.1 with 32 bytes of data:
Reply from 192.168.0.1: bytes=32 time=37ms TTL=255
Reply from 192.168.0.1: bytes=32 time=11ms TTL=255
Reply from 192.168.0.1: bytes=32 time=14ms TTL=255
Reply from 192.168.0.1: bytes=32 time=28ms TTL=255
Ping statistics for 192.168.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 11ms, Maximum = 37ms, Average = 22ms
C:\>
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
WRT300N
Wireless-Router0
PC-PT
PC0
PC-PT
PC2
PC-PT
PC1
Simulation

Experiment 8(b):
Configuration of wireless LAN using Cisco Packet Tracker

Aim:
To configure a WLAN using cisco packet tracker by connecting three PCs to a Linksys Wireless Router, assigning static IP Address, configuring the SSID and WEP security, and verifying connectivity.

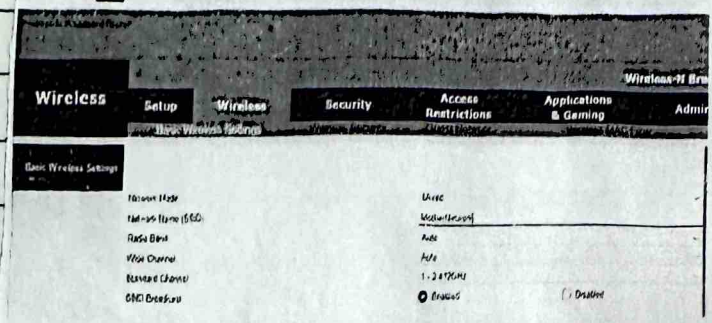
Procedure:

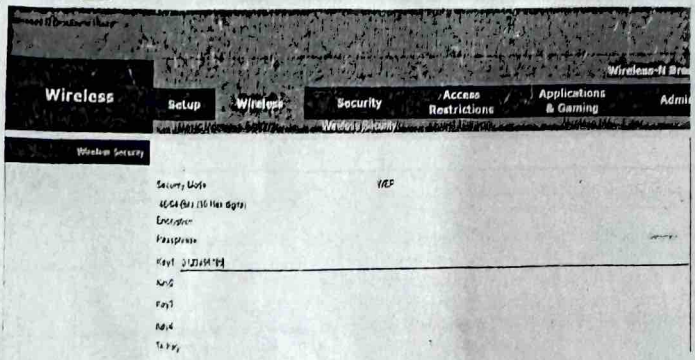
1. Open Cisco Packet Tracker and place a Linksys Wireless Router and PCs in the workspace.

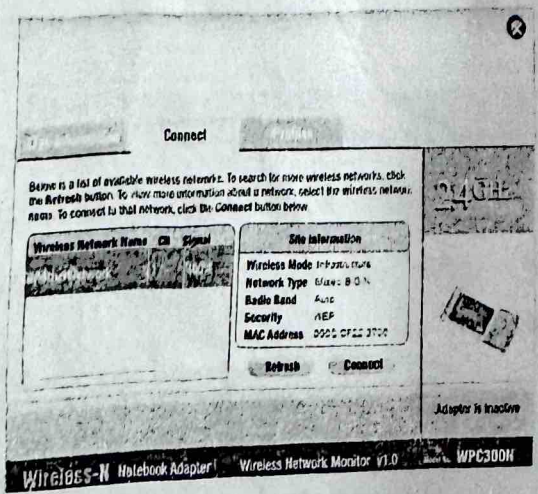
2. Open the wireless Router and go the administration tab, set the username and password as admin, then save the settings.

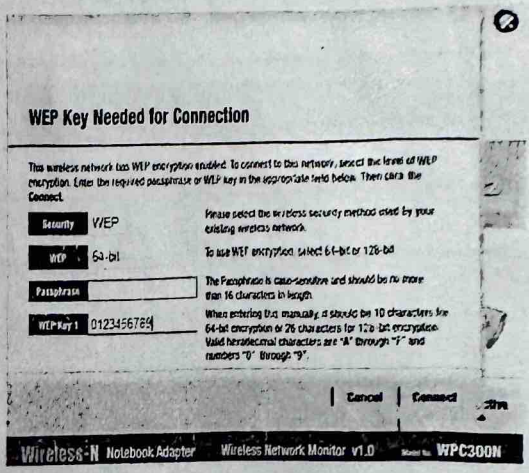
3. Go to the wireless tab and change the the default SSID to:
MotherNetwork

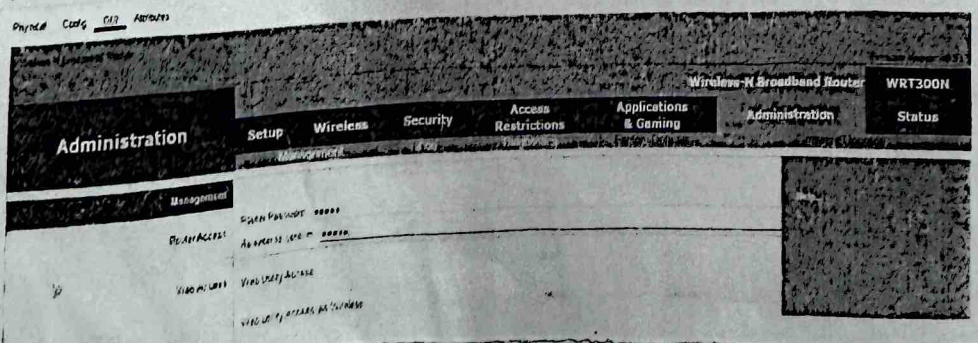
4. Open Wireless Security, select WEP as the security mode and set:
Key 1: 0123456789


Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Basic Wireless Settings


Wireless
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Wireless Security


Connect
Below is a list of available wireless networks. To search for more wireless networks, click the Refresh button. To view more information about a network, select the wireless network name. To connect to that network, click the Connect button below.
Wireless Network Name
Site Information
Wireless Mode
Network Type
Radio Band
Security
MAC Address
Refresh
Connect
Adapter is inactive
Wireless-N Notebook Adapter
Wireless Network Monitor V1.0
WPC300N


WEP Key Needed for Connection
This wireless network uses WEP encryption enabled. To connect to this network, select the level of WEP encryption. Enter the required passphrase or WEP key in the appropriate fields below. Then click the Connect.
Security
WEP
Please select the wireless security method used by your existing wireless network.
WEP
64-bit
To use WEP encryption, select 64-bit or 128-bit.
Passphrase
The Passphrase is case-sensitive and should be no more than 16 characters in length.
WEP Key 1
0123456789
When entering this manually, it should be 10 characters for 64-bit encryption or 26 characters for 128-bit encryption. Valid hexadecimal characters are "A" through "F" and numbers "0" through "9".
Cancel
Connect
Wireless-N Notebook Adapter
Wireless Network Monitor V1.0
WPC300N


Administration
Setup
Wireless
Security
Access Restrictions
Applications & Gaming
Administration
Status
Wireless-N Broadband Router
WRT300N
Management

5. Configure the static IP address of the three PCs as follows:

PC	IP Address	Subnet Mask	Default Gateway
PC_0	192.168.0.2	255.255.255.0	192.168.0.1
PC_1	192.168.0.3	255.255.255.0	192.168.0.1
PC_2	192.168.0.4	255.255.255.0	192.168.0.1

6. For each PC, Open Desktop → PC Wireless → Connect and refresh the available networks

7. Select MotherNetwork and click connect

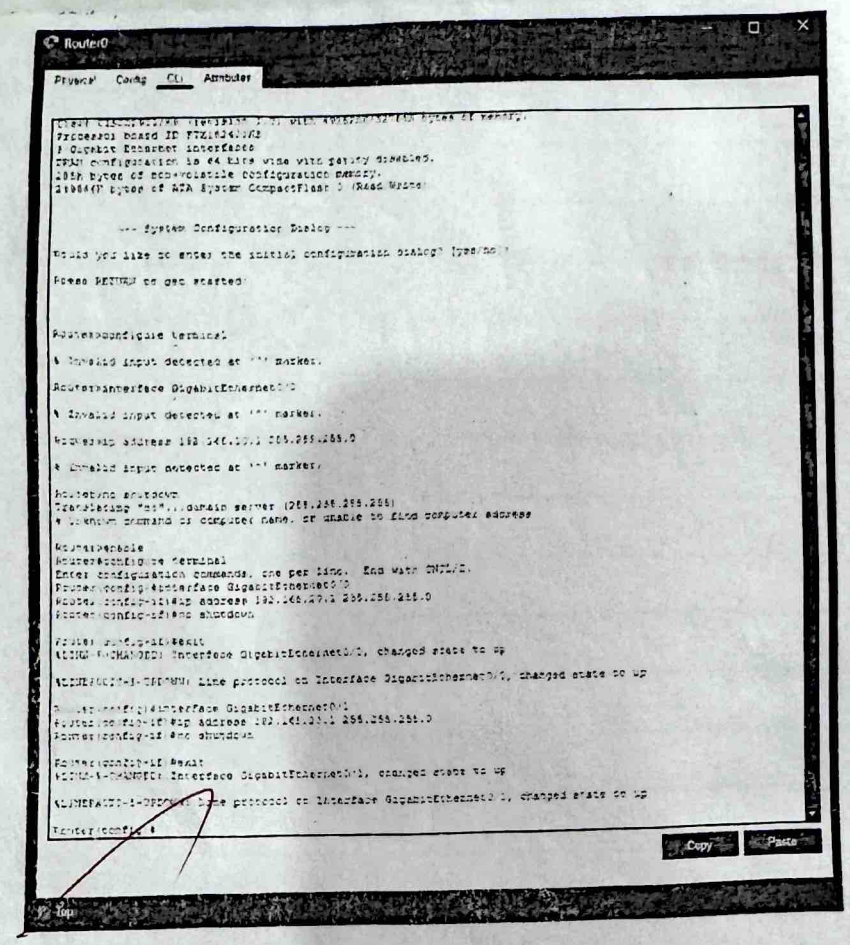
8. Enter the WEP key:
   0123456789

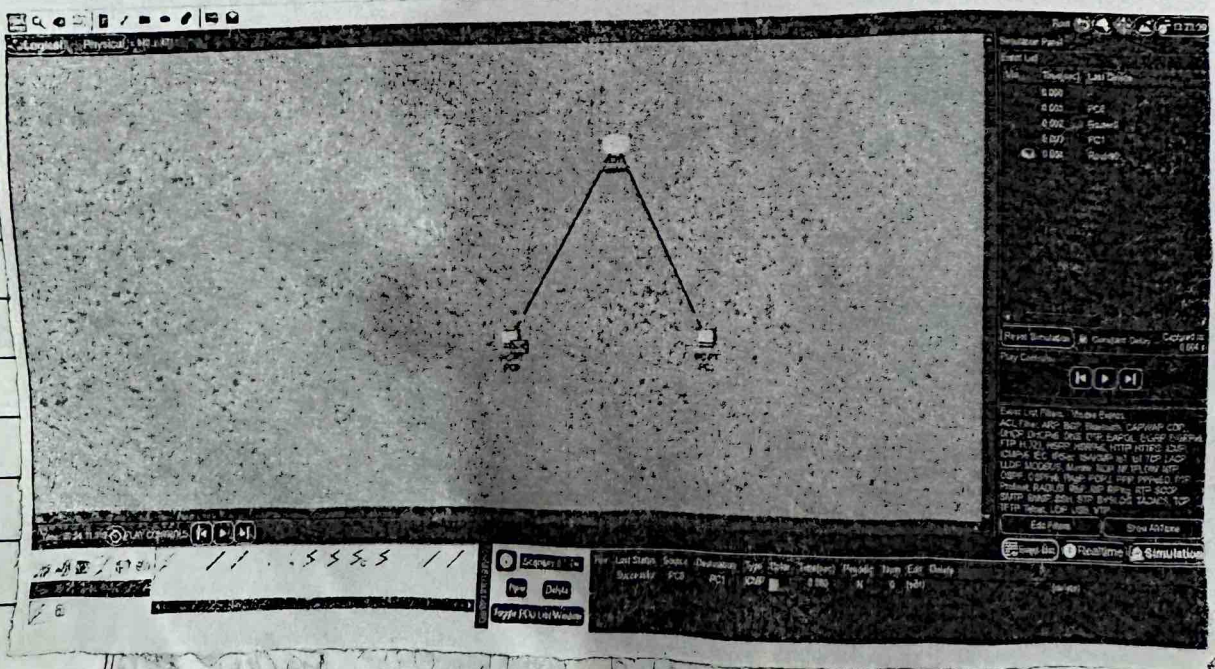
9. Repeat the same wireless connection process for PC_1 and PC_2

10. Open Command Prompt on PC_0 and Verify connectivity using
    ping 192.168.0.3
    ping 192.168.0.4
    ping 192.168.0.1

11. Verify that all ping request recive replies with 0% packet loss

Result:
The wireless LAN was successfully configured using Cisco packet Tracker. The Three PCs were connected to Linksys Wireless Router using Mother Network SSID and WEP security. Verified using the ping command with 0% packet loss.


Router0
Physical
Config
CLI
Attributes
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]:
Press RETURN to get started!
Router>enable
Router#configure terminal
% Invalid input detected at '^' marker.
Router#interface GigabitEthernet0/0
% Invalid input detected at '^' marker.
Router#ip address 192.168.10.1 255.255.255.0
% Invalid input detected at '^' marker.
Router#no shutdown
Translating "shutdown"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.30.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
Router(config)#
Copy
Paste


Physical
PC0
PC1
Scenario 0
Fire
Delete
Toggle PDU List Window
Realtime
Simulation

Experiment - 10

Internetworking with Routers

Aim:

To design and configure a simple internetwork using a router in Cisco Packet Tracker simulator and simulate packet transmission between two different network.

Procedure:

1. Open Cisco Packet tracer and create a network topology.
2. Add the following devices
   * 1 Router - Router 0
   * 2 PCs - PC0 and PC1 and PC1
3. Connect PC0 to Router 0 using Copper Straight Through cable.
4. Open Router0 → CLI and enter privileged mode:
   enable
   configure terminal
5. Configure the first and second routers interface:
   interface GigabitEthernet 0/0
   ip address 192.168.10.1 255.255.255.0
   no shutdown
   exit

6. Configure PC0 using Desktop → IP Configuration
   * IP Address : 192.168.10.2
   * Subnet Mask : 255.255.255.0
   * Default Gateway: 192.168.10.1

7. Configure PC1 :
   * IP Address 192.168.20.2
   * Subnet Mask : 255.255.255.0
   * Default Gateway 192.168.20.1

8. Switch to simulation mode

9. Select Add Simple PDU and click PC0 as the source and PC1 as the destination

10. Use Capture/Forward to trace the packet

11. Observe the packet path :
    PC0 → Router0 → PC1

12. Continue the simulation
    PC1 → Router0 → PC0

13. Verify the successful status and tick mark on PC0, confirming that communication between the two networks was successful.

Result :
Thus, a simple internetwork was successfully designed and configured using a router in Cisco Packet Tracker. Packet transmission between PC0 and PC1 through the router was successfully simulated and verified.

Exp No: 10b

Wireless Networking

Aim:

To design and configure an internetwork using a wireless router, DHCP server and Internet cloud.

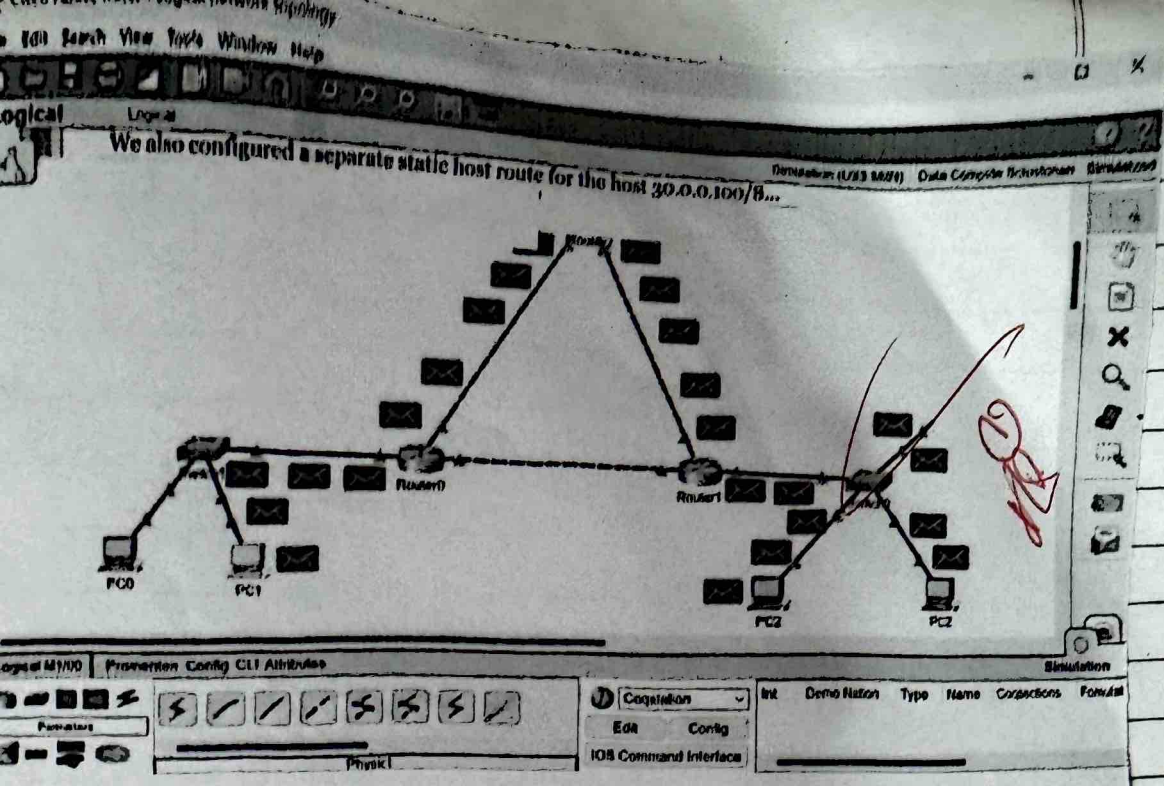
Algorithm:

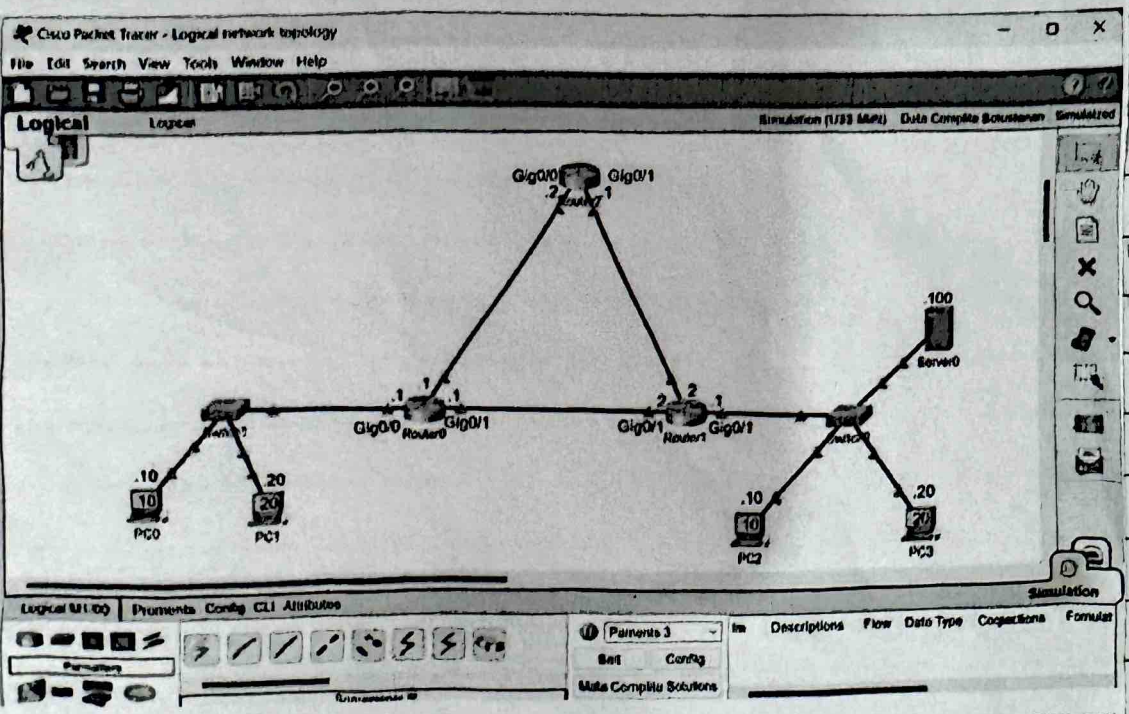
1. Open Cisco Packet Tracker
2. Add a wireless router, DHCP server and internet cloud.
3. Add required wireless devices
4. Connect the devices to the wireless router
5. Configure wireless router settings
6. Configure DHCP server with IP settings
7. Enable DHCP for automatic IP assignment
8. Connect the wireless devices to the network
9. Verify the assignment IP address
10. Test connectivity between devices and Internet cloud.

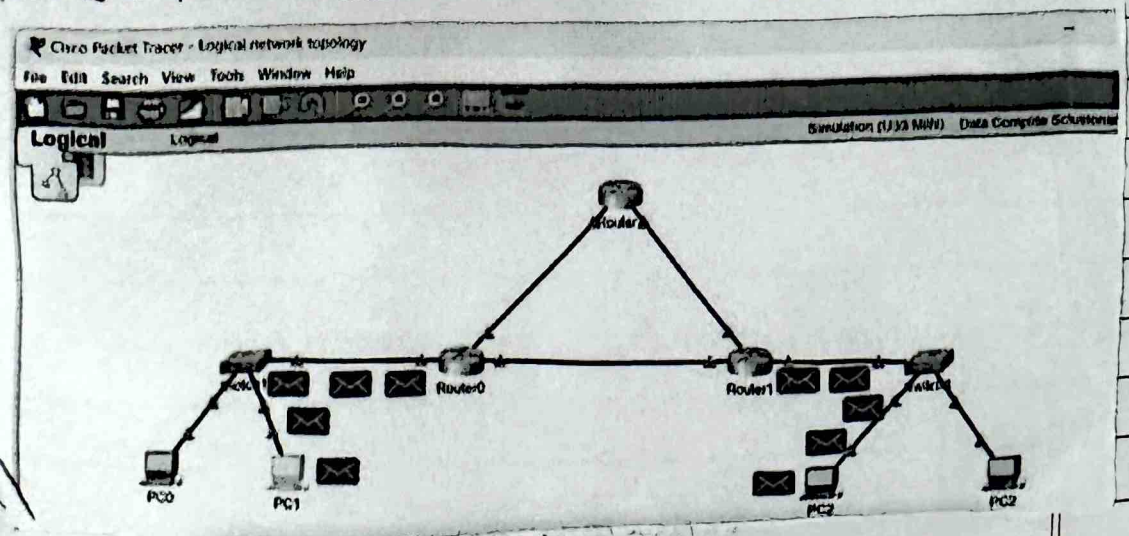
Result:

Thus, the wireless internetwork was successfully configured and connectivity was verified.

We also configured a separate static host route for the host 30.0.0.100/8...







Exp no: 11a

Date
Page

Static Routing

Aim:

To configure static routing using Cisco packet tracer

Algorithm:

1. Open Cisco Packet Tracer
2. Create the required router and PC topology.
3. Connect the devices using suitable cables.
4. Assign IP addresses to PC and router interface.
5. Enable router interfaces using no shutdown.
6. Configure static routes using IP route
7. Check the routing table using show IP route.
8. Test connectivity using ping
9. Verify communication between networks.

Result:

Thus, the echo client server was successfully implemented using sockets.

Exp No: 11b

Date
Page

RIP Routing

Aim:

To configure RIP routing using Cisco Packet Tracer

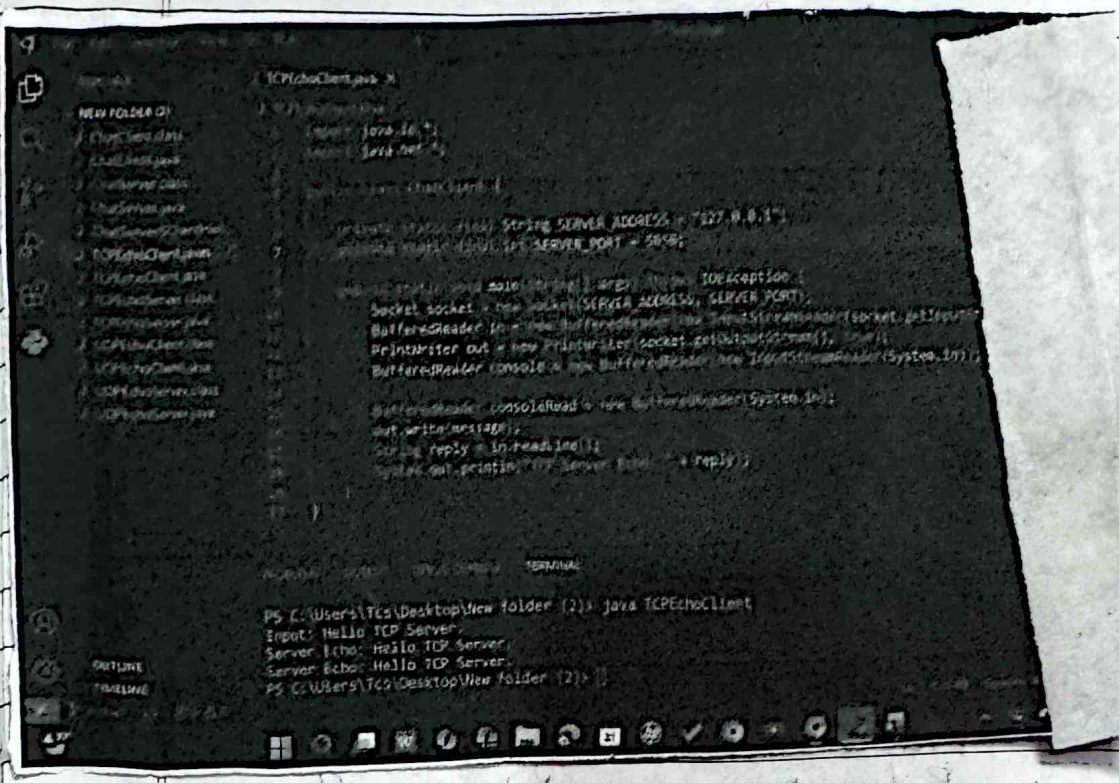
Algorithm:

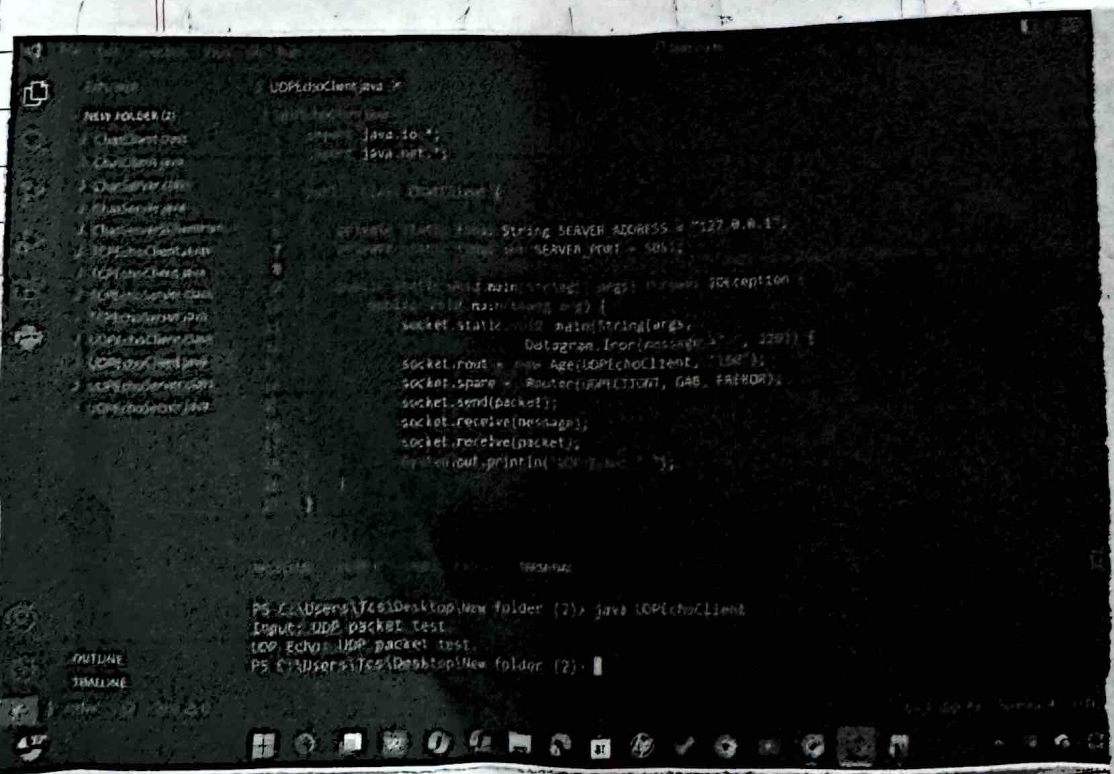
1. Open Cisco Packet Tracer
2. Create the required router and PC topology
3. Connect the devices using suitable cables.
4. Assign IP addresses to PCs and router interface.
5. Enable router interface using no shutdown
6. Enter RIP configuration mode using router rip.
7. Add connected networks using network commands.
8. Configure RIP on required router
9. Check the routing table
10. Test connectivity using ping.

Result:

Thus, RIP routing was successfully configured and connectivity between the networks was verified.

Date
Page


TCPEchoClient.java
NEW FOLDER (2)
import java.io.*;
import java.net.*;
String SERVER_ADDRESS = "127.0.0.1";
int SERVER_PORT = 5056;
IOException
Socket socket = new Socket(SERVER_ADDRESS, SERVER_PORT);
PrintWriter out = new PrintWriter(socket.getOutputStream(), true);
BufferedReader console = new BufferedReader(new InputStreamReader(System.in));
out.println(message);
String reply = in.readLine();
System.out.println("Server Echo: " + reply);
TERMINAL
PS C:\Users\TCS\Desktop\New folder (2)> java TCPEchoClient
Input: Hello TCP Server.
Server Echo: Hello TCP Server.
Server Echo: Hello TCP Server.
PS C:\Users\TCS\Desktop\New folder (2)>
OUTLINE
TIMELINE


UDPEchoClient.java
NEW FOLDER (2)
import java.io.*;
import java.net.*;
String SERVER_ADDRESS = "127.0.0.1";
SERVER_PORT = 5056;
socket.send(packet);
socket.receive(message);
socket.receive(packet);
TERMINAL
PS C:\Users\TCS\Desktop\New folder (2)> java UDPEchoClient
Input: UDP packet test.
UDP Echo: UDP packet test.
PS C:\Users\TCS\Desktop\New folder (2)>
OUTLINE
TIMELINE

Exp No: 12(a)

Date
Page

Echo Client-Server

Aim:

To implement an echo client-server using TCP/UDP sockets.

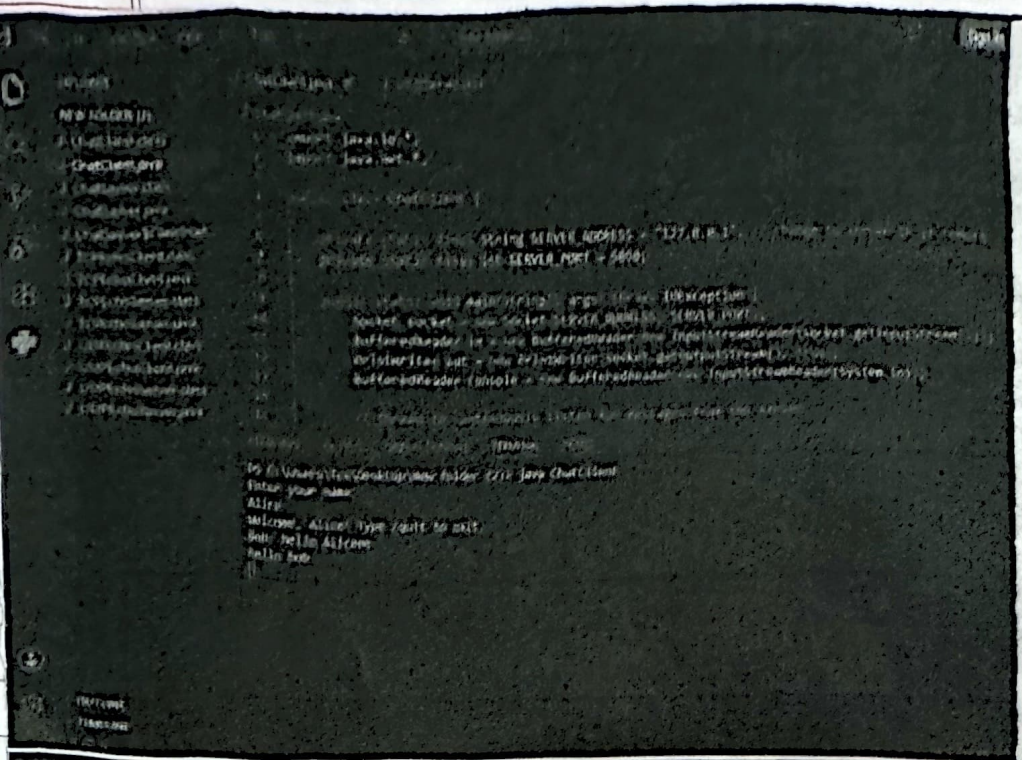
Algorithm:

* Create the server socket
* Bind the socket to an IP address and port.
* Create the client socket
* Connect the client to the server
* Enter a message from the client
* Send the message to the server
* Server recives and echoes the message
* Client recives the echoed message
* Close the sockets

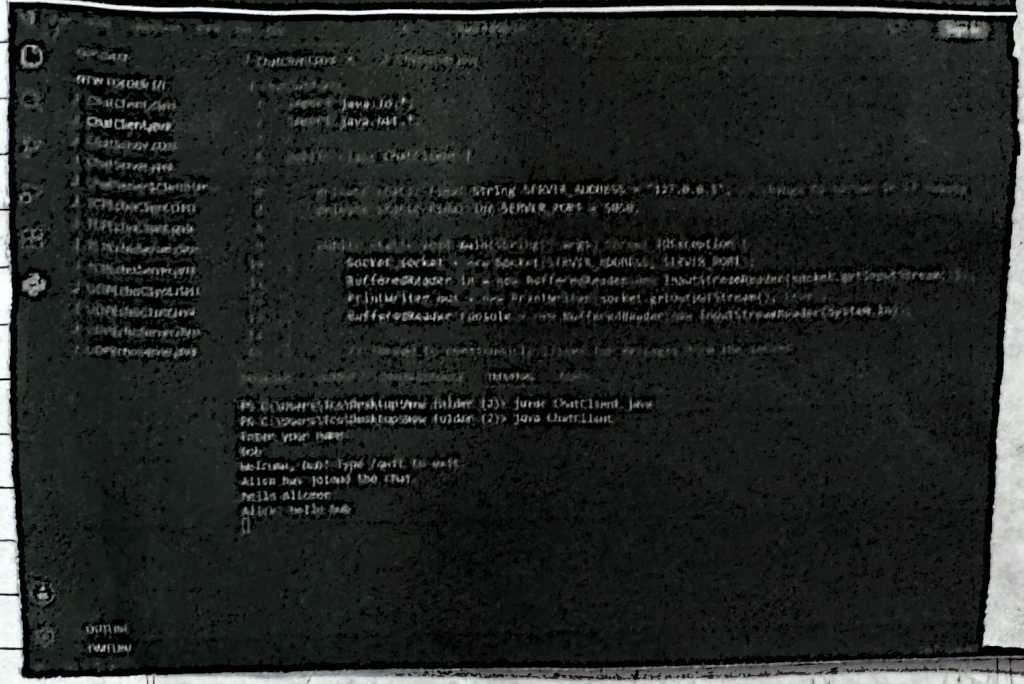
Result:

Thus, echo client-server was succesfully implemented using socket

Date
Page



```
Enter your name:
Alice
Welcome, Alice! Type /quit to exit.
Bob: hello Alice
Hello Bob
```



```
PS C:\Users\Ira\Desktop\New folder (2)> javac ChatClient.java
PS C:\Users\Ira\Desktop\New folder (2)> java ChatClient
Enter your name:
Bob
Welcome, Bob! Type /quit to exit.
Alice has joined the chat
Hello Alice!!
Alice: hello bob
```

88

Exp No: 12(b)

Date
Page

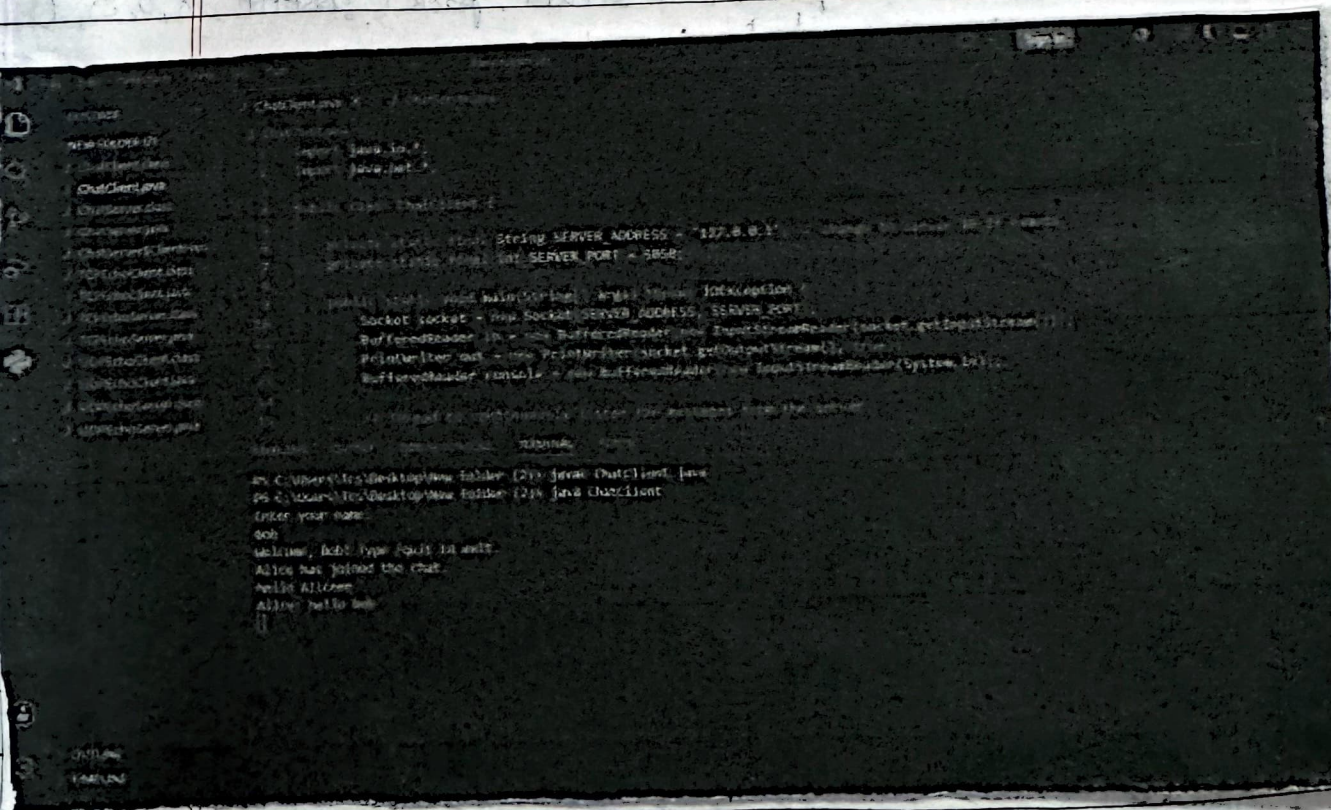
CHAT - CLIENT SERVER

Aim:

To implement a chat program using socket programming.

Algorithm:

1. Create Server Socket
2. Bind the socket to IP address and port.
3. Start listening for client.
4. Establish connection between client and server
5. Enter a message from client
6. Send and Recive message between client and server
7. Display recived message
8. Continue communication until termination
9. Close Socket.



Exp No:13

Date
Page

Implement Your Own Ping Program

Aim: To implement a ping program using python raw sockets to check host reachability

Code:

```
import socket, struct, time, os
def checksum(data):
    total = sum((data[i] << 8) + (data[i+1]
                for i in range(0, len(data), 2)))
    total = (total >> 16) + (total & 0xffff)
    return (~total) & 0xffff
def ping(host):
    try: addr = socket.gethostbyname(host)
        s = socket.socket(socket.AF_INET, socket.
        s.settimeout(2)  IPPROTO_ICMP)
        pid = os.getpid() & 0xffff
        header = struck.pack("!BHHH", 8,0,0,pi
        data = b"PING"
        cs = checksum(header+data)
        packet = struct.pack("!BBHHH", 8,0,c
        start = time.time()
        s.sendto(packet, (addr, 1))
        s.recvfrom(1024)
        print("Relpy from", addr, "Time",
        sc.close()
    except socket.timeout:
        print("Request timeout")
    except OSError:
        print("Permission denied or network
ping(input("Enter destination host:"))
```

Sample I/P and O/P

Enter destination host : 8.8.8.8

Reply from 8.8.8.8. time = 24.53ms

Result:

Thus, the ping program was succesfully implemented in Python.

Exp No: 14

Date
Page

Packet Sniffing Using Raw Sockets

Aim:

To capture network packets using python raw sockets and display source IP, destination IP and protocol.

Code:

```
import socket, struct
s = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
print("Packet Sniffer Started...")
while True:
    packet, addr = s.recvfrom(65535)
    ip = struct.unpack("!BBHHHBBH4s4s", packet[:20])
    src = socket.inet_ntoa(ip[8])
    dst = socket.inet_ntoa(ip[9])
    print("Source IP:", src)
    print("Destination IP:", dst)
    print("Protocol:", ip[6])
    print("Packet Size:", len(packet), "bytes")
    print("-" * 40)
```

O/P

```
Packet Sniffer Started...
Source IP: 192.168.1.10
Destination IP: 142.250.188.4
Protocol: 6
Packet Size: 60 bytes
----------------------------------------
Source IP: 8.8.8.8
Destination IP: 192.168.1.10
Protocol: 1
Packet size: 84 bytes
```

Result: Thus, ping program and packet sniffing program were sucessfully

Exp no: 15

Date
Page

Analysis of Various Types of Servers using Webalizer

Aim: To analyze web server log files using weblizer and study request, visit, pages, transfered data and HTTP status codes

Code:

```
from collections import Counter
with open("access.log", "r") as f:
    logs = f.readlines()
status = Counter()
ips = set()
pages = set()
total bytes = 0
for log in logs:
    parts = log.split()
    if len(parts) > 6:
        ips.add(parts[0])
        pages.add(parts[6])
        status[parts[8]] += 1
        try:
            total bytes += int(parts[9])
        except ValueError:
            pass
print("Total Requests:", len(logs))
print("Total Visitors:", len(ips))
print("Total Pages:", len(pages))
print("Total Data:", total_bytes, "bytes")
print("HTTP status Codes:", dict(status))
```

Output:

Total Request : 1250
Total visitors : 480
Total Pages : 920
Total Data : 15400000 bytes
HTTP Status Codes : {'200' : 1180, '404' : 45, '500' : 25}

Result:

Thus, the webalizer tool was used successfully to analyse webserver log files and various server statistics such as requests, visitors, pages, bandwith and HTTP responses code were obtained.

Algorithm:

* Start Webalizer Application
* Install and Configure webalizer
* Configure webalizer with required log file path
* Run Webalizer on the selected log file
* Webalizer reads and analyzes the server log entries
* It generates statistics such as:
    * Number of visits
    * Number of requests
    * Pages accessed
* Open the generated HTML report in web browser
* Analyse server activity.